

**Unleashing the Potential of Chat GPT:
Enhancing Chronic Disease Management in the healthcare sector
Bachelor thesis**

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List of abbreviations

ADRD – Alzheimer’s disease and related dementias

ADKS – Alzheimers Disease Knowledge Scale

AI - artificial intelligence

API – Application Programming Interface

BERT – bidirectional encoder representations from transformers

Bi-daily – twice a day

Chat GPT – chat generative pre-trained transformer

Chat GPT-4 – chat generative pre-trained transformer version 4

Cf. -

Ebd. – ebendas

Et al –

EU – European Union

FDA – Food and Drug Administration

FKS - Flesch-Kincaid Grade Level Score

FM MHS – The Functional Medicine Master Health Survey

FMP - Family Medicine Practitioner

FMP’s – Functional Medicine Practitioners

FRS - Flesch Reading Score

GBD – global burden of disease

GDP – Gross Domestic Product

IgE – immunoglobulins

JSON – JavaScript Object Notation

LLMs – large language models

LLMS – Linear Least Mean Square

MD5 – Message-Digest Algorithm 5

ML – machine learning

NEMT – nonemergency medical transportation

NLP – natural language processing

NoSQL – Not only Structured Query Language

PLWD – Persons Living with Disabilities

QA – question-answering

rLHF

SD – Standard deviation

SSL – Secure Sockets Layer

SSRIs – selective serotonin reuptake inhibitors

URLs – Uniform Resource Locator

U.S. – United States

USA – United States of America

Vs. – versus

WHO – World Health Organization

QA – Question-answering

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1. Introduction to Chat GPT in chronic disease Management

1.1 Understanding chronic disease management in the healthcare sector

Chronic diseases pose a significant challenge to the healthcare sector worldwide. Conditions such as diabetes, hypertension, asthma and heart disease require long-term management and can have a profound impact on individuals' quality of life. In recent years, there has been a growing interest in leveraging advanced technologies to enhance chronic disease management and one such technology that holds great promise is Chat GPT.¹

Chat GPT, powered by the latest advanced in artificial intelligence, has the potential to revolutionize the way individuals with chronic conditions receive support and care. This sub chapter aims to explore how Chat GPT can assist healthcare providers in offering personalized advice, monitoring symptoms and providing lifestyle recommendations to those managing chronic diseases.²

One of the key benefits of Chat GPT is its ability to offer personalized advice tailored to the needs of each individual. By analyzing vast amounts of medical data and patient history, Chat GPT can provide accurate and timely recommendations for medication adjustments, dietary modifications, and exercise routines. This personalized approach takes into account the unique circumstances of each patient, ensuring that the advice given is both effective and relevant.³

In addition to personalized advice, Chat GPT can also play vital role in monitoring symptoms. Patients can interact with Chat GPT through various platforms, such as mobile apps or web interfaces, to report their symptoms and receive real-time feedback. This enables healthcare providers to track disease progression more effectively and intervene promptly when needed. Patients can also benefit from continuous monitoring, as Chat GPT can alert them to any changes or abnormalities that may require immediate attention.⁴

¹ Cf. Schachner T, Keller R, v Wangenheim F., Artificial Intelligence-Based Conversational Agents for Chronic Conditions: Systematic Literature Review J. Med Internet Res 2020;22(9):e20701, <https://www.jmir.org/2020/9/e20701/PDF>, Zugriff am 15.09.2023.

² Ebd.

³ Ebd.

⁴ Ebd.

Furthermore, Chat GPT can offer lifestyle recommendations to individuals with chronic conditions. Managing a chronic disease often entails making significant lifestyle changes, such as adopting healthier diet, engaging in regular physical activity, and managing stress levels. Chat GPT can provide personalized suggestions and reminders, helping patients adhere to these lifestyle modifications and improve their overall well-being.⁵

In conclusion, Chat GPT has the potential to revolutionize chronic disease management in the healthcare sector. By offering personalized advice, monitoring symptoms, and providing lifestyle recommendations, Chat GPT can support individuals with chronic conditions in their journey towards better health. As the technology continues to evolve, it is crucial for healthcare providers to embrace and utilize Chat GPT to enhance patient care and outcomes in the field of chronic disease management.⁶

1.2 The role of technology in healthcare

In recent years, the healthcare sector has witnessed a rapid evolution in the way technology is being used to improve patient care and outcomes. From electronic medical records to telemedicine, advancements in technology have revolutionized various aspects of healthcare delivery. One area where technology has immense potential is chronic disease management. This subchapter explores the role of technology, specifically Chat GPT, in enhancing chronic disease management in the healthcare sector.⁷

Chronic diseases, such as diabetes, heart disease, and asthma, affect millions of individuals worldwide. Managing these conditions requires constant monitoring of symptoms, adherence to treatment plans, and making necessary lifestyle changes. However, traditional healthcare models often fall short in providing personalized and continuous support to individuals with chronic conditions. This is where Chat GPT comes into play.⁸

Chat GPT, an advanced conversational AI model, has the capability to offer personalized advice, monitor symptoms, and provide lifestyle recommendations to

⁵ Cf. Haque, Ahshanul; Chowdhury, Md Naseef-Ur-Rahman; Soliman, Hamdy (2023). Transforming Chronic Disease Management with Chatbots: Key Use Cases for Personalized and Cost-effective Care. TechRxiv. Preprint. <https://doi.org/10.36227/techrxiv.22579735.v2>, accessed on 15.09.2023.

⁶ Ebd.

⁷ Cf. Benjamin Chaix, Arthur Guillemassé, Pierre Nectoux, et al. Vik: A Chatbot to Support Patients, <https://biomedres.us/pdfs/BJSTR.MS.ID.004453.pdf>, accessed on 15.09.2023.

⁸ Ebd.

individuals with chronic diseases. By leveraging natural language processing and machine learning, Chat GPT can understand and respond to patients' queries, concerns, and symptoms in real-time.⁹

One of the best key benefits of Chat GPT in chronic disease management is the ability to offer personalized advice. By analyzing patient data, including medical history, symptoms, and treatment plans, Chat GPT can provide tailored recommendations to individuals. For instance, a patient with diabetes can receive personalized dietary advice and insulin dosage recommendations based on their glucose levels and lifestyle factors. This personalized approach not only improves patient engagement but also enhances treatment outcomes.¹⁰

Furthermore, Chat GPT can continuously monitor patients' symptoms and provide interventions. For instance, individuals with asthma can receive alerts and reminders to take their medications or use their inhaler based on their reported symptoms or environmental triggers. By closely monitoring symptoms, Chat GPT can help individuals manage their conditions effectively and prevent exacerbations.¹¹

Lastly, Chat GPT can offer lifestyle recommendations to individuals with chronic diseases. For instance, patients with heart diseases can receive personalized exercise routines and stress management techniques to improve their overall cardiovascular health. By incorporating machine learning algorithms, Chat GPT can even learn from patient feedback and continuously refine its recommendations, ensuring the advice provided is up-to-date and evidence based.¹²

In conclusion, technology, particularly Chat GPT, holds immense potential in enhancing chronic disease management in the healthcare sector. By offering personalized advice, monitoring symptoms, and providing lifestyle recommendations, Chat GPT can support individuals with chronic conditions in their journey towards better health. As the healthcare sector continues to embrace technology advancements, the integration of Chat GPT in chronic disease

⁹ Cf. Bin Sawad, A.; Narayan, B.; Alnefaie, A.; Maqbool, A.; Mckie, I.; Smith, J.; Yuksel, B.; Puthal, D.; Prasad, M.; Kocaballi, A.B. A Systematic Review on Healthcare Artificial Intelligent Conversational Agents for Chronic Conditions. *Sensors* 2022, 22, 2625, https://mdpi-res.com/d_attachment/sensors/sensors-22-02625/article_deploy/sensors-22-02625-v2.pdf?version=1648701343, accessed on 14.09.2023.

¹⁰ Cf. Bin Sawad, A.; Narayan, B.; Alnefaie, A.; Maqbool, A.; Mckie, I.; Smith, J.; Yuksel, B.; Puthal, D.; Prasad, M.; Kocaballi, A.B. A Systematic Review on Healthcare Artificial Intelligent Conversational Agents for Chronic Conditions. *Sensors* 2022, 22, 2625, https://mdpi-res.com/d_attachment/sensors/sensors-22-02625/article_deploy/sensors-22-02625-v2.pdf?version=1648701343, accessed on 14.09.2023.

¹¹ Ebd.

¹² Ebd.

management has the potential to revolutionize part care and improve outcomes in the long run.¹³

1.3 Introduction to Chat GPT and natural language processing

In recent years, the healthcare sector has witnessed remarkable advancements in technology, with the potential to transform the way we manage chronic diseases. One such breakthrough in the development of Chat GPT, a cutting-edge natural language processing (NLP) model that has shown great promise in enhancing chronic disease management.¹⁴

Chronic disease management is a complex and ongoing process that requires constant monitoring, personalized advice, and lifestyle recommendations. With the advent of Chat GPT, healthcare professionals now have a powerful tool at their disposal to support individuals with chronic conditions in a more effective and tailored manner.¹⁵

Chat GPT utilizes NLP, a branch of artificial intelligence, to understand and generate human-like text responses. This makes it an ideal tool for engaging in natural and meaningful conversations with individuals seeking support for managing their chronic conditions.¹⁶

One of the key benefits of Chat GPT in chronic disease management is its ability to offer personalized advice. By analyzing an individual's medical history, symptoms, and lifestyle factors, Chat GPT can provide tailored recommendations that align with their specific needs. This personalized approach not only promotes better adherence to treatment plans but also empowers individuals to take an active role in managing their conditions.¹⁷

Moreover, Chat GPT can assume a pivotal function in the surveillance of symptoms. By regularly interacting with individuals and asking targeted questions, Chat GPT is capable of acquiring significant data regarding the advancement of symptoms, their intensity, and prospective catalysts. Healthcare

¹³ Cf. Schachner T, Keller R, von Wangenheim F
Artificial Intelligence-Based Conversational Agents for Chronic Conditions: Systematic Literature Review
J Med Internet Res 2020;22(9):e20701, <http://www.jmir.org/2020/9/e20701/>, accessed on 20.08.2023.

¹⁴ Cf. Schachner T, Keller R, von Wangenheim F
Artificial Intelligence-Based Conversational Agents for Chronic Conditions: Systematic Literature Review
J Med Internet Res 2020;22(9):e20701, <http://www.jmir.org/2020/9/e20701/>, accessed on 20.08.2023.

¹⁵ Ebd.

¹⁶ Ebd.

¹⁷ Cf. Chowdhury, Md Naseef-Ur-Rahman; Haque, Ahshanul; Soliman, Hamdy (2023). Chatbots: A Game Changer in mHealth. TechRxiv. Preprint. <https://doi.org/10.36227/techrxiv.22548532.v2>, accessed on 28.08.2023.

providers may utilize this current data to modify treatment plans and treatments as necessary, so guaranteeing the most effective management of the condition.¹⁸ In addition to offering customized advice and symptom monitoring, Chat GPT possesses the capability to furnish lifestyle recommendations. This tool has the potential to aid individuals in making more health-conscious decisions pertaining to their dietary habits, physical activity, stress reduction, and adherence to prescribed medications. Through the introduction of positive lifestyle modifications, Chat GPT has the potential to substantially enhance the general state of well-being and quality of life for persons afflicted with chronic diseases.¹⁹ The ongoing development of Chat GPT and NLP in the field holds great promise for improving chronic illness management within the healthcare industry. Through the utilization of artificial intelligence and natural language processing, doctors are now able to provide tailored, timely, and evidence-based assistance to persons suffering from chronic ailments. This advancement holds the potential to enhance results and transform the approach to chronic illness management.²⁰

2. The potential of Chat GPT in chronic disease management

2.1 Personalized advice and support for individuals with chronic conditions

In today's healthcare sector, managing chronic conditions has become a significant challenge. The increasing prevalence of chronic diseases, such as diabetes, heart disease and asthma, calls for innovative solutions that can offer personalized advice and support to individuals facing these conditions. This sub chapter delves into the potential of Chat GPT in enhancing chronic disease management by investigating how it can provide personalized advice, monitor symptoms and offer lifestyle recommendations to individuals with chronic conditions.²¹

One of the key benefits of Chat GPT is its ability to provide personalized advice to individuals with chronic conditions. By leveraging vast amounts of medical data, the AI-powered chat bot can analyze an individual's health history, symptoms and treatment plans to offer tailored recommendations. For instance,

¹⁸ Ebd.

¹⁹ Cf. Chowdhury, Md Naseef-Ur-Rahman; Haque, Ahshanul; Soliman, Hamdy (2023). Chatbots: A Game Changer in mHealth. TechRxiv. Preprint. <https://doi.org/10.36227/techrxiv.22548532.v2>, accessed on 28.08.2023.

²⁰ Ebd.

²¹ Cf. Lou J, Wang C, Liu S: Utility of ChatGPT in Clinical Practice (2023), <https://www.jmir.org/2023/1/e48568/PDF>, accessed on 18.09.2023.

someone with diabetes can receive personalized advice on managing their blood sugar levels, diet and exercise routine. This personalized approach helps individuals feel supported and empowers them to take control of their health.²² In addition, Chat GPT can play a crucial role in monitoring symptoms and providing real-time support. The chatbot can engage in conversational exchanges with individuals, asking about their symptoms, severity and frequency. This information can be used to assess the individual's condition and track any changes over time. In case of any concerning symptoms, the chatbot can alert healthcare professionals, ensuring timely interventions and preventing potential complications.²³

Moreover, to advice and symptom monitoring, Chat GPT can offer lifestyle recommendations to individuals with chronic conditions. It can provide guidance on healthy habits, stress management techniques and self-care practices that can improve overall well-being. For example, individuals with heart disease can receive recommendations into their daily lives, individuals can better manage their conditions and improve their quality of life.²⁴

The potential of Chat GPT in chronic disease management its immense. By offering personalized advice, monitoring symptoms and providing lifestyle recommendations, AI-powered chatbot can fill the gaps in traditional healthcare systems. They can provide continuous support, empower individuals to take charge of their health and enhance overall management outcomes.²⁵

As healthcare sector continues to evolve, it is vital to explore innovative solutions like Chat GPT that can revolutionize chronic disease management. By harnessing the power of AI, we can unleash the potential of personalized advice and support for individuals with chronic conditions, ultimately improving their health and well-being.²⁶

2.2 Monitoring symptoms and progression of chronic diseases

In the realm of chronic disease management, continuous monitoring of symptoms and progression plays a crucial role in improving patient outcomes and enhancing

²² Ebd.

²³ Cf. Lou J, Wang C, Liu S: Utility of ChatGPT in Clinical Practice (2023), <https://www.jmir.org/2023/1/e48568/PDF>, accessed on 18.09.2023.

²⁴ Ebd.

²⁵ Ebd.

²⁶ Ebd.

their quality of life traditionally, this monitoring process has relied heavily on in-person consultations with healthcare professionals, which can be time-consuming, expensive and inconvenient for patients. However, with the advent of advanced technology like Chat GPT, the landscape of chronic disease management is poised to undergo a dramatic shift.²⁷

Chat GPT, with its remarkable ability to understand and generate human-like text, can revolutionize the way individuals with chronic conditions receive support and guidance. Through the use of NLP, machine learning, and extensive medical databases, Chat GPT has the capacity to provide individualized guidance, track symptoms, and make lifestyle suggestions that are specifically targeted to the distinct requirements of the patient.²⁸

One of the primary advantages of Chat GPT is its ability to engage in real-time conversations with patients. Through chat interfaces or voice-enabled applications, individuals can easily communicate their symptoms, concerns and questions, allowing Chat GPT to provide immediate responses and guidance. This real-time interaction not only enhances patient experience, but also timely interventions can be made to prevent complications and manage chronic conditions effectively.²⁹

Meanwhile, Chat GPT can continuously monitor symptoms by keeping track of patient-reported data and analyzing it against established medical guidelines. By monitoring symptoms such as pain levels, fatigue or shortness of breath, Chat GPT can detect patterns, identify triggers and provide insights into disease progression. This information can be shared with healthcare providers, facilitating more informed decision-making and enabling interventions to be tailored to the patient's evolving needs.³⁰

Similarly, to symptom monitoring, Chat GPT can offer personalized lifestyle recommendations to individuals with chronic conditions. By leveraging patient-reported data, medical history, and evidence-based guidelines, Chat GPT can provide advice on exercise routines, dietary modifications, stress management techniques, and medication adherence. These tailored recommendations empower

²⁷ Cf. Lasorsa I, D'Antrassi P, Ajčević M, Stellato K, Di Lenarda A, Marceglia S, Accardo A. Personalized support for chronic conditions: a novel approach for enhancing self-management and improving lifestyle. *Appl Clin Inform* 2016; 7: 633–645 <http://dx.doi.org/10.4338/ACI-2016-01-RA-0011>, accessed on 18.09.2023.

²⁸ C. Lasorsa I, D'Antrassi P, Ajčević M, Stellato K, Di Lenarda A, Marceglia S, Accardo A. Personalized support for chronic conditions: a novel approach for enhancing self-management and improving lifestyle. *Appl Clin Inform* 2016; 7: 633–645 <http://dx.doi.org/10.4338/ACI-2016-01-RA-0011>, accessed on 18.09.2023.

²⁹ Ebd.

³⁰ Ebd.

patients to take an active role in managing their health, leading to improved self-care and overall well-being.³¹

To summarize the preceding discussion, the integration of Chat GPT into chronic disease management holds immense potential for supporting individuals with chronic conditions. By providing personalized advice, monitoring symptoms in real-time, and offering lifestyle recommendations, Chat GPT can transform the way healthcare professionals and patients collaborate in managing chronic diseases. This technology has the power to enhance patient outcomes, promote self-management, and ultimately revolutionize the healthcare sector's approach to chronic disease management.³²

2.3 Providing lifestyle recommendations for better disease management

In the realm of chronic disease management, it's imperative to explore innovative solutions that can enhance the lives of individuals with chronic conditions. One such solution that holds immense potential is Chat GPT, a cutting-edge technology that can revolutionize the healthcare sector. By offering personalized advice, monitoring symptoms, and providing lifestyle recommendations, Chat GPT has the power to transform the landscape of chronic disease management.³³ When it comes to managing chronic conditions effectively, lifestyle modifications play a crucial role. By making small yet impactful changes in their daily routines, individuals can significantly improve their overall well-being and mitigate the symptoms of their conditions. Chat GPT acts as a virtual companion, guiding individuals towards these lifestyle changes, and providing them with recommendations tailored to their specific needs.³⁴ One key aspect that Chat GPT excels in is personalized advice. By leveraging advanced algorithms and machine learning capabilities, Chat GPT can analyze an individual's health data, symptoms, and medical history to offer personalized recommendations. Whether it's dietary changes, exercise routines, stress management techniques, or sleep patterns, Chat GPT can provide tailored advice that empowers individuals to take control of their health.³⁵

³¹ Cf. Tustumi F, Andreollo NA, Aguilar-Nascimento JE. ABCD Arq Bras Cir Dig. 2023;36e1727, <https://doi.org/10.1590/0102-672020230002e1727>, accessed on 18.09.2023.

³² Cf. Lasorsa I, D'Antrassi P, Ajčević M, Stellato K, Di Lenarda A, Marcegaglia S, Accardo A. Personalized support for chronic conditions: a novel approach for enhancing self-management and improving lifestyle. Appl Clin Inform 2016; 7: 633–645 <http://dx.doi.org/10.4338/ACI-2016-01-RA-0011>, accessed on 18.09.2023.

³³ Ebd.

³⁴ Ebd.

³⁵ Ebd.

Chat GPT's ability to monitor symptoms in real-time adds an extra layer of support. By regularly communicating with individuals, Chat GPT can gather information about their symptoms, track their progress, and alert healthcare providers if any concerning patterns emerge. This proactive approach ensures early intervention, preventing exacerbation of symptoms and reducing hospital admissions.³⁶

Besides that, Chat GPT can assist individuals in setting realistic goals and milestones, keeping them motivated throughout their chronic disease management journey. By providing continuous feedback, encouragement, and reminders. It becomes a trusted ally, helping individuals stay on track and make sustainable lifestyle changes.³⁷

In conclusion, the integration of Chat GPT into chronic disease management offers immense potential for individuals with chronic conditions. By offering personalized advice, monitoring symptoms, and providing lifestyle recommendations, Chat GPT can empower individuals to lead healthier lives.³⁸

3. Exploring the capabilities of Chat GPT for chronic disease management

3.1 Natural Language Processing and its application in healthcare

Natural Language Processing (NLP) is a field of artificial intelligence (AI) that focuses on the interaction between computers and human language. It has gained significant attention in recent years due to its potential applications in various industries, including healthcare. In this subchapter, we will explore how NLP can be leveraged in the healthcare sector, specifically in the context of chronic disease management.³⁹

Chronic diseases, such as diabetes, cardiovascular diseases, and respiratory disorders, are significant burden on individuals and healthcare systems. They require long-term management, personalized care, and continuous monitoring of symptoms and lifestyle factors. This is where NLP can play a crucial role.⁴⁰

³⁶ Ebd.

³⁷ Cf. Lasorsa I, D'Antrassi P, Ajčević M, Stellato K, Di Lenarda A, Marceglia S, Accardo A. Personalized support for chronic conditions: a novel approach for enhancing self-management and improving lifestyle. *Appl Clin Inform* 2016; 7: 633–645 <http://dx.doi.org/10.4338/ACI-2016-01-RA-0011>, accessed on 18.09.2023.

³⁸ Ebd.

³⁹ Cf. Locke, Saskia et al. "Natural language processing in medicine: A review." *Trends in Anaesthesia and Critical Care* (2021): n. pag., <https://doi.org/10.1016/j.tacc.2021.02.007>, accessed on 06.04.2024.

⁴⁰ Cf. Haque, Ahshanul et al. "Transforming Chronic Disease Management with Chatbots: Key Use Cases for Personalized and Cost-effective Care." *2023 Sixth International Symposium on Computer, Consumer and Control (IS3C)* (2023): 367-370, <https://doi.org/10.1109/IS3C57901.2023.00104>, 15.04.2024.

One of the primary applications of NLP in chronic disease management is the development of chatbots powered by AI, such as Chat GPT. These chatbots can provide personalized advice, monitor symptoms, and offer lifestyle recommendations with chronic conditions.⁴¹

By analyzing large volumes of medical literature, patient records, and treatment guidelines, NLP algorithms can extract relevant information and generate context-specific responses. This enables chatbots to provide accurate and up-to-date information to individuals, empowering them to make informed decisions about their health.⁴²

NLP algorithms can understand and process unstructured text, such as electronic health records and patient-generated data. This allows chatbots to monitor symptoms reported by individuals, identify potential exacerbations or complications, and provide timely interventions or recommendations. For example, a chatbot can detect changes in blood glucose levels in a diabetic patient and suggest appropriate dietary modifications or medication adjustments.⁴³

In addition to symptom monitoring, NLP can support lifestyle recommendations by analyzing individuals' preferences, habits, and social determinants of health. Chatbots can offer personalized advice on exercise routines, dietary choices, stress management techniques, and medication adherence. By tailoring recommendations to each individual's unique needs, chatbots can help achieve better health outcomes and improve their quality of life.⁴⁴

Overall, NLP holds immense potential for enhancing chronic disease management in the healthcare sector. By leveraging AI-powered chatbots like Chat GPT, healthcare professionals can extend their reach, provide personalized care, and empower individuals to take an active role in managing their chronic conditions.⁴⁵ This subchapter aims to investigate the various ways in which NLP can support individuals with chronic diseases, ultimately improving their well-being and reducing the burden on healthcare systems.

3.2 Training Chat GPT to understand and respond to chronic disease queries

⁴¹ Ebd.

⁴² Cf. Ogbuju, Emeka and Georgina Nkolika Obunadike. "Information Extraction from Electronic Medical Records using Natural Language Processing Techniques." *Journal of Applied Sciences and Environmental Management* (2020): n. pag., <https://doi.org/10.4314/jasem.v24i6.13>, accessed on 06.05.2024.

⁴³ Cf. Montagna, Sara et al. "Data Decentralisation of LLM-Based Chatbot Systems in Chronic Disease Self-Management." *Proceedings of the 2023 ACM Conference on Information Technology for Social Good* (2023): n. pag., <https://doi.org/10.1145/3582515.3609536>, accessed on 06.05.2024.

⁴⁴ Ebd.

⁴⁵ Ebd.

In recent times, the healthcare industry has experienced significant technological developments, easing to a transformative impact on the management of chronic illnesses. One notable breakthrough is the introduction of Chat GTP, a robust tool that possesses significant promise for augmenting the management of chronic diseases. Chat GPT can serve as a virtual assistant for persons with chronic diseases by providing tailored advice, monitoring symptoms, and delivering lifestyle suggestions. This enables users to enhance their quality of life and assume greater agency over their health.⁴⁶

To fully leverage the capabilities of Chat GPT, it is required to train it in understanding and responding to chronic disease requests. By implementing this approach, healthcare professionals may effectively secure the integration of this AI-driven technology as a very important resource in their repertoire facilitating the provision of individualized treatment to patients afflicted with chronic ailments.⁴⁷

The process of acquiring knowledge, skills, and competencies through systematic instruction and practice. The process of utilizing Chat GTP entails providing it with extensive quantities of data pertaining to chronic ic illnesses, encompassing symptoms, available treatment alternatives, modifications to lifestyle, and accounts from patients. Through the process of data analysis, Chat GPT can acquire the ability to see trend and develop a complete comprehension of diverse chronic diseases for which it is specifically meant to provide assistance.⁴⁸

After undergoing training, Chat GPT has the capability to offer individualized recommendations that are customized to meet the specific requirements of every patient. For example, in the case of a diabetic patient seeking advice on the management of their blood glucose levels, Chat GPT can provide suggestions pertaining to dietary choices, physical activity, and modifications to drug regimes, all tailored to the specific circumstances of the person. Through the provision of real-time and individualized advice, Chat GPT has the potential to let consumers to make well-informed decisions regarding their health.⁴⁹

⁴⁶ Cf. Tariqa, Saba and Sundus Tariqb. "Revolutionizing Healthcare with AI: The Role of ChatGPT." *Journal of University Medical & Dental College* (2023): n. pag., <https://doi.org/10.37723/jumdc.v14i2.850>, accessed on 06.05.2024.

⁴⁷ Cf. Ray, Partha Pratim and Poulami Majumder. "The Potential of ChatGPT to Transform Healthcare and Address Ethical Challenges in Artificial Intelligence-Driven Medicine." *Journal of Clinical Neurology (Seoul, Korea)* 19 (2023): 509 – 511, <https://doi.org/10.3988/jcn.2023.0158>, accessed on 06.05.2024.

⁴⁸ Cf. Yang, H. et al. "Exploring the Potential of Large Language Models in Personalized Diabetes Treatment Strategies." *medRxiv* (2023), <https://doi.org/10.1101/2023.06.30.23292034>, accessed on 06.05.2024.

⁴⁹ Cf. Sharma, Samriddhi et al. "A Critical Review of ChatGPT as a Potential Substitute for Diabetes Educators." *Cureus* 15 (2023): n. pag., <https://doi.org/10.7759/cureus.38380>, accessed on 25.03.2024.

In addition, Chat GPT has the capability to observe symptoms and offer prompt interventions. In the event that a patient diagnosed with asthma communicates a heightened occurrence of wheezing and difficulty in breathing, Chat GPT has the capability to advise the patient to adhere to their recommended medication regimen or, if warranted, to promptly seek medical intervention. The use of a proactive strategy for monitoring symptoms has the potential to mitigate exacerbations and enhance the treatment of the condition.⁵⁰

In conjunction with guidance and the monitoring of symptoms, Chat GPT has the capability to provide lifestyle suggestions to persons who are afflicted with chronic diseases. Through the examination of data pertaining on the influence of several lifestyle elements on the advancement of diseases, Chat GPT has the capability to propose alterations encompassing stress mitigation methods, practices promoting healthy sleep patterns, and tactics aimed at discontinuing smoking habits. These ideas have the potential to facilitate the adoption of healthier behaviors and enhance persons' overall well-being.⁵¹

The utilization of Chat GTP in the management of chronic diseases has the potential to significantly transform the delivery of healthcare services, hence revolutionizing the manner in which persons with chronic disorders are provided care. The rapid progression of technology necessitates healthcare practitioners to actively adopt and investigate these advancements in order to optimize the management of chronic diseases.⁵²

3.3 Challenges and limitations of implementing Chat GPT in healthcare

The healthcare industry is actively exploring novel technologies to improve the management of chronic diseases, and Chat GPT has emerged as a viable tool in this regard. Nevertheless, it is important to acknowledge and address several obstacles and constraints that must be taken into account prior to the complete integration of this technology within the healthcare industry.⁵³

One of the foremost obstacles is in the task of guaranteeing the precision and dependability of the information disseminated by Chat GPT. Although the model

⁵⁰ Cf. Joshi, Ashish et al. "Clinical Impact of Home Automated Telemanagement in Asthma." *AMIA ... Annual Symposium proceedings. AMIA Symposium* (2005): 1000, <https://api.semanticscholar.org/CorpusID:7294264>, accessed on 06.05.2024.

⁵¹ Cf. Montagna, Sara et al. "Data Decentralisation of LLM-Based Chatbot Systems in Chronic Disease Self-Management." *Proceedings of the 2023 ACM Conference on Information Technology for Social Good* (2023): n. pag., <https://doi.org/10.1145/3582515.3609536>, accessed on 06.05.2024.

⁵² Ebd.

⁵³ Cf. Ali, Hazrat et al. "ChatGPT and Large Language Models in Healthcare: Opportunities and Risks." *2023 IEEE International Conference on Artificial Intelligence, Blockchain, and Internet of Things (AIBThings)* (2023): 1-4, <https://doi.org/10.1109/AIBThings58340.2023.10291020>, accessed on 06.05.2024.

possesses the capability to create replies by leveraging extensive datasets, there exists a potential for the generation of erroneous or misleading information. It is imperative for healthcare practitioners to use caution in validating the advice and suggestions offered by Chat GPT in order to safeguard patient well-being and mitigate any potential risks.⁵⁴

Another significant difficulty that arises in this context pertains to the preservation of patient privacy and the safeguarding of data security. The Chat GPT platform engages in personalized interactions with users, wherein it may gather health-related data of a sensitive nature. Establishing effective data protection protocols is of utmost importance in order to preserve sensitive information and ensure compliance with privacy legislation. In order to mitigate the risk of unauthorized access and data breaches, it is imperative for healthcare companies to allocate resources towards the establishment of robust infrastructure and the use of encryption solutions.⁵⁵

In addition, it is imperative to overcome the limits of Chat GPT in comprehending intricate medical vocabulary and contextual nuances. The system may have challenges in comprehending subtle symptoms or effectively evaluating the extent of a medical issue. Enhancing the accuracy and efficacy of the model may be achieved by training it on comprehensive medical datasets and integrating domain-specific expertise.⁵⁶

Furthermore, a notable shortcoming of Chat GPT lies in its deficiency of human touch and empathy in its exchanges. Individuals diagnosed with chronic illnesses frequently want emotional assistance and affirmation, posing a potential obstacle for chatbots utilizing artificial intelligence to fulfill such needs. Healthcare practitioners should explore strategies for incorporating human intervention when deemed essential in order to promote patients' sense of understanding and support.⁵⁷

Furthermore, it is important to take into account the accessibility and acceptance of Chat GPT. Certain individuals, particularly elderly persons or those with poor computer literacy, may experience discomfort when utilizing such equipment. In order to achieve widespread acceptance and optimize the benefits of Chat GPT,

⁵⁴ Ebd.

⁵⁵ Cf. Kuang, Yi-Rui et al. "ChatGPT encounters multiple opportunities and challenges in neurosurgery." *International Journal of Surgery (London, England)* 109 (2023): 2886 – 2891, <https://doi.org/10.1097/JS9.0000000000000571>, accessed on 06.05.2024.

⁵⁶ Ebd.

⁵⁷ Cf. Hopkins, Ashley M. et al. "Artificial intelligence chatbots will revolutionize how cancer patients access information: ChatGPT represents a paradigm-shift." *JNCI Cancer Spectrum* 7 (2023): n. pag., <https://doi.org/10.1093/jncics/pkad010>, accessed on 06.05.2024.

healthcare institutions should prioritize the provision of user-friendly interfaces and adequate training.⁵⁸

In summary, it is evident that Chat GPT possesses significant promise in augmenting the management of chronic diseases. However, it is crucial to acknowledge and tackle many obstacles and constraints before its extensive use within the healthcare industry. It is imperative to examine several critical issues, including the assurance of information correctness and dependability, the preservation of patient privacy and data security, the resolution of limits in medical knowledge, the incorporation of human empathy, and the facilitation of accessibility and acceptance. By proactively confronting these obstacles, healthcare practitioners may utilize the capabilities of Chat GPT to provide tailored assistance and enhance the well-being of those suffering from chronic ailments.⁵⁹

4. The potential in healthcare of Chat GPT with Bard and other KI's

This chapter aims to comprehensively explore the possible uses and limits of Chat GPT and other AI techniques within the healthcare sector.

In the year 2023, Dave offers a comprehensive analysis of the benefits, constraints, and ethical implications of Chat GPT in the field of medicine. This analysis emphasizes the potential utility of Chat GPT in areas such as clinical diagnosis, medical education, and virtual patient support.⁶⁰

The primary objective of Wu 2023 is to enhance the performance of Chat GPT within the medical domain by the integration of medical information and the facilitation of few-shot learning. This research showcases the potential of Chat GPT as a flexible assistant.⁶¹

The study conducted by Gary in 2023 investigates the use of Chat GPT in the context of patient care and medical research. The research emphasizes the potential of Chat GPT in several aspects, including patient queries, decision support, and research assistance. However, the study also acknowledges the

⁵⁸ Ebd.

⁵⁹ Cf. Hopkins, Ashley M. et al. "Artificial intelligence chatbots will revolutionize how cancer patients access information: ChatGPT represents a paradigm-shift." *JNCI Cancer Spectrum* 7 (2023): n. pag., <https://doi.org/10.1093/jncics/pkad010>, accessed on 06.05.2024.

⁶⁰ Cf. Ethics guidelines for trustworthy AI | Shaping Europe's digital future. (2019), <https://digital-strategy.ec.europa.eu/en/library/ethics-guidelines-trustworthy-ai> (accessed March 6, 2023), accessed on 04.10.2023.

⁶¹ Cf. Qualifying Chinese Medical Licensing Examination with Knowledge Enhanced Generative Pre-training Model, Jiageng Wu, X. Wu, Zhaopeng Qiu, Mighui Li, Yemeni Zheng, Joe Yang, <https://doi.org/10.48550/arXiv.2305.10163>, accessed on 04.10.2023.

existence of problems pertaining to the accuracy, bias and legal implications associated with the use of Chat GPT in these domains.⁶²

In the final analysis, Ramachandran (2023) examines the prospective influence of Chat GPT inside the realm of medical education, with particular emphasis on the advantages it presents to both students and tutors. However, the author also raises apprehensions regarding compromised integrity and excessive dependence on this technology.⁶³

In general, the before mentioned articles indicate that Chat GPT and similar AI technologies has the capacity to augment healthcare practices. Nevertheless, it is essential to exercise caution and deliberate over the ethical and practical ramifications associated with their use.

4.1 A comparative analysis of search result: Chat GPT vs. Google search engine for inquiries pertaining to dementia and cognitive decline

Individuals who are affected by dementia or other forms of cognitive decline, as well as their carers, are increasingly turning to the internet as a primary source for accessing information pertaining to their disease, as well as locating relevant resources and assistance. The current progressions in the field of large language models (LLMs), exemplified as Chat GPT, present a novel alternative to conventional online search engines, such as Google.⁶⁴

Web search engines, such as Google and Bing, engage in the ongoing process of collecting material from the internet, known as crawling. This content is stored locally and indexed to facilitate quicker retrieval when users submit queries. Upon the submission of a query by a user, the search engine employs a ranking algorithm to assign a numerical score to each page, therefore generating a list of results in a ranked fashion.⁶⁵

In recent times, search engines have been endeavoring to transition from the conventional keyword-based search approach, wherein users input keywords and receive a list of relevant web pages, to a QA paradigm. In this new approach, users pose questions, such as “What is the most widely prescribed medication for

⁶² Cf. Garg, Ravindra Kumar et al. “Exploring the Role of Chat GPT in patient care (diagnosis and Treatment) and medical research: A Systematic Review.” *Health Promotion Perspectives* (2023), <https://doi.org/10.1101/2023.06.13.23291311>, accessed on 04.10.2023.

⁶³ Cf. Ramachandran, Vignesh et al. “A Tête-à-tête with ChatGPT on the impact of artificial intelligence in medical education.” *The Medical journal of Malaysia* 78 4 (2023), <https://api.semanticscholar.org/CorpusID:260333221>, accessed on 04.10.2023.

⁶⁴ Cf. Hristidis V, Ruggiano N, Brown EL, Ganta SRR, Stewart S. ChatGPT vs Google for Queries Related to Dementia and Other Cognitive Decline: Comparison of Results *J Med Internet Res* 2023;25:e48966, <https://www.jmir.org/2023/1/e48966>, accessed on 04.10.2023.

⁶⁵ Cf. Mark Levene, Search Engines: Information Retrieval in Practice, *The Computer Journal*, Volume 54, Issue 5, May 2011, Pages 831–832, <https://doi.org/10.1093/comjnl/bxq039>, accessed on 04.10.2023.

depression?” and the search engine provides a direct answer, such as “SSRIs”, prominently displayed at the top of the search result page. This is followed by the customary list of web pages for further exploration.⁶⁶

The specific algorithm employed by web search engines to determine rankings remains undisclosed to the public; nevertheless, the broad methodologies employed are well recognized. Prior to 2018, the ranking process used many factors, including the frequency of query terms inside an article, the presence of inbound hyperlinks to the content, and the significance of the domain.

The aforementioned features are integrated through a function that has been acquired through the process of machine learning.⁶⁷

In 2018, a significant advancement occurred with the publication of a research article by Google on bidirectional encoder representations from transformers (BERT). BERT is a deep learning model specifically designed for text analysis. BERT is a language model that generates a semantic representation of a given text segment, such as a word, phrase, or paragraph. This semantic representation enables the comparison of the text segment to a query in terms of meaning and context. Several notable advantages of BERT in comparison to prior keyword-based approaches are outlined below:

- 1) Synonyms are regarded as viable options, or more broadly, alternative means of conveying identical significance;
- 2) the entirety of the text is taken into account rather than solely emphasizing pivotal words, such as nouns;
- 3) the arrangement of words assumes significance;
- 4) the precise section of a document that addresses the query is provided instead of presenting the entire document in its entirety.

The use of BERT has resulted in notable enhancements in the overall efficacy of search outcomes, hence facilitating a closer alignment of online search engines with the objective of achieving question-answering capabilities.⁶⁸

ChatGPT is an interactive conversational system, sometimes referred to as a chatbot, designed to provide responses to user inquiries in a manner that closely resembles human interaction.⁶⁹

⁶⁶ Cf. Dense Passage Retrieval for Open-Domain Question Answering, Karpukhin et al., EMNLP 2020, <https://aclanthology.org/2020.emnlp-main.550>, accessed on 04.10.2023.

⁶⁷ Cf. Mark Levene, Search Engines: Information Retrieval in Practice, *The Computer Journal*, Volume 54, Issue 5, May 2011, Pages 831–832, <https://doi.org/10.1093/comjnl/bxq039>, accessed on 04.10.2023.

⁶⁸ Cf. Devlin, Jacob et al. “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding.” *North American Chapter of the Association for Computational Linguistics*(2019), <https://api.semanticscholar.org/CorpusID:52967399>, accessed on 04.10.2023.

⁶⁹ Cf. Introducing ChatGPT. OpenAI. 2022 Nov 30., <https://openai.com/blog/chatgpt/>, accessed on 04.10.2023.

ChatGPT is constructed utilizing the third-generation Generative Pre-trained Transformer (GPT-3), a language model that has undergone training with about 500 billion tokens (terms or words). The majority of these tokens are derived from online sites, specifically a subset of pages that have been indexed by a web search engine, such as Google. Additionally, tokens are also obtained from Wikipedia and books. Significantly, the training of language models incurs substantial costs, resulting in few training sessions, often occurring once per year.

Consequently, these models may not encompass the most up-to-date information that is now accessible.⁷⁰

A language model may be conceptualized as a computational tool that is capable of generating coherent and plausible extensions for a given section of text. As an illustration, in the given text ‘it is sunny in,’ a language model may propose extending the phrase with the geographical location of ‘California’. One notable enhancement of Chat GPT in comparison to GPT-3 is in its focus on generating chatbot replies that are more user-oriented and meaningful. Specifically, Chat GPT aims to provide responses that address the user’s inquiries, as opposed to merely generating text that would typically follow the users input in a document. In order to accomplish this objective, Chat GPT undergoes training, specifically fine-tuning, in conjunction with the existing training of GPT-3. This process incorporates human feedback with the aim of enhancing the use of Chat GPT beyond that of GPT-3. Chat GPT is fine-tuned through the utilization of reinforcement learning from human input. This process enables the model to enhance its following responses by evaluating the perceived utility of its answers to the user at hand.⁷¹

The retraining of Chat GPT occurs at infrequent intervals, typically spanning many months or even years, mostly due to the substantial expenses associated with the training process. Consequently, the responses generated by Chat GPT may not reflect the most current information when referencing recent occurrences. The Chat GPT website provides users with the opportunity to access Chat GPT. In order to gain access to the system, new users are required to register by providing their email address. Users are allocated a limited quantity of credits, which are

⁷⁰ Cf. Brown, Tom B. et al. “Language Models are Few-Shot Learners.”; <https://api.semanticscholar.org/CorpusID:218971783>, accessed on 04.10.2023.

⁷¹ Cf. Ouyang, Long et al. “Training language models to follow instructions with human feedback.”; <https://api.semanticscholar.org/CorpusID:246426909>, accessed on 04.10.2023.

afterwards utilized to cover the per usage cost. This fee is determined by the number of tokens that are sent to Chat GPT.⁷²

4.2 Investigation: Comparing the efficacy of Chat GPT and Google for inquiries regarding dementia and cognitive decline

A systematic approach was devised for the collecting processing of data. The reasearch encompassed the identification of frequently asked topics pertaining to ADRD and caring by caregivers. Additionally, a set of criteria was established to evaluate the replies obtained via Chat GPT and Google. Further elaboration on this procedure will be provided thereafter.

4.2.1 Identification of the issue

Two sorts of questions typically asked by carers were identified:

Informative questions and transactional questions, which align with the previously proposed categorization. Because of the factors that were stated in the introduction section, researchers took into consideration two of the three kinds of web inquiries.⁷³

In order to addresses general information inquiries, scientists made alterations to specific items desired from the ADKS. The ADKS is a rigorously tested evaluation instrument comprising 30 items pertaining to Alzheimer disease. Participants are required to indicate th veracity of each statement by selecting either ‘true’ or ‘false’. The statements encompass several subjects pertaining to evaluation and diagnosis, caregiving, course of the condition, influence on daily life, treatment and management strategies, and symptoms. As an illustration, let us examine the subsequent two statements:

- 1) Individuals diagnosed with Alzheimers disease exhibit an elevated propensity for experiencing falls as the progrsssion of the disease advances.
- 2) The presence of hypertension has been associated with an elevated likelihood of acquiring Alzheimer’s disease.⁷⁴

⁷² Cf. Welcome to ChatGPT. ChatGPT, <https://chat.openai.com/>, accessed on 04.10.2023.

⁷³ Cf. Andrei Broder. 2002. A taxonomy of web search. SIGIR Forum 36, 2 (Fall 2002), 3–10. <https://doi.org/10.1145/792550.792552>, accessed on 04.10.2023.

⁷⁴ Cf. Carpenter BD, Balsis S, Otilingam PG, Hanson PK, Gatz M. The Alzheimer's Disease Knowledge Scale: development and psychometric properties. Gerontologist 2009 Apr;49(2):236-247, <https://europepmc.org/abstract/MED/19363018>, accessed on 04.10.2023.

In order to transform these assertions into interrogative form, researchers have incorporated the prefix ‘Is it true that’ into each sentence. For instance, ‘Is it true that individuals afflicted with Alzheimer’s disease exhibit an elevated propensity to experience falls as the condition progresses?’. The ADKS is well recognized as a prominent instrument for evaluating individuals’ comprehension of Alzheimer’s disease and its associated concepts. The knowledge questions including several areas were created with the purpose of identifying deficiencies in the knowledge of patients and caregivers, particularly those who are seeking a dementia evaluation and have concerns linked to dementia. As a result, this tool is suitable for those who are in search of information.⁷⁵

The study utilized a set of 30 transactional items that are routinely employed in the provision of services and support for individuals with dementia. Families and caregivers faced with the responsibility of caring for a family member suffering from dementia frequently find themselves in the position of needing to acquire knowledge regarding the identification of reliable, high-quality, accessible, and cost-effective services. The following list comprises 30 different types of services that are relevant to dementia care. The focal inquiries are to the manner in which Persons Living with Disabilities (PLWD) seek for the necessary resources within their immediate geographical vicinity.⁷⁶ The list was reduced to six prevalent kinds of services, namely “adult day-, home health-, hospice- and respite care, memory clinics and nonemergency medical transportation (NEMT).”⁷⁷ Concerns regarding the quality, accessibility, and pricing of each of these services were raised by individuals. In particular, a set of five questions were formulated for each sort of service. As an illustration, shown below is a compilation of five inquiries formulated for adult day care:

- How do I pick the best adult day care? (Quality)
- How do I find adult day care in Riverside, California? (Accessibility)
- How do I pay for adult day care in California? (Affordability)
- How much does adult day care cost per day in Riverside, California? (Affordability)
- Does California license adult day care? (Quality)”⁷⁸

⁷⁵ Ebd.

⁷⁶ Cf. Brian D. Carpenter, Steve Balsis, Poorni G. Otilingam, Priya K. Hanson, Margaret Gatz, The Alzheimer’s Disease Knowledge Scale: Development and Psychometric Properties, *The Gerontologist*, Volume 49, Issue 2, April 2009, Pages 236–247, <https://doi.org/10.1093/geront/gnp023>, accessed on 14.10.2023.

⁷⁷ Cf. Alzheimers.gov. Finding Dementia Care and local Services, <https://www.alzheimers.gov/life-with-dementia/find-local-services>, accessed on 14.10.2023.

⁷⁸ Hristidis V, Ruggiano N, Brown EL, Ganta SRR, Stewart S
ChatGPT vs Google for Queries Related to Dementia and Other Cognitive Decline: Comparison of Results J Med Internet Res 2023;25:e48966, <https://www.jmir.org/2023/1/e48966>, accessed on 14.10.2023.

The identical inquiries were formulated for each of the remaining five categories of services. The place chosen for the inquiries was Riverside, California. Therefore, a cumulative sum of 30 transactional inquiries was recorded. The comprehensive compilation of all 60 questions is displayed in the following. The supplementary material provided in this study is referred to as Multimedia Appendix 1.⁷⁹

4.2.2 The process of gathering and organizing data

One researcher, referred to as author SG, conducted the analysis by inputting each query into the Chat GPT and Google search engines from March 10 to 13, 2023, utilizing a computer situated in Riverside, California. Author VH provided supervision and oversight for team member SG during the collecting procedure. Feedback on the process was offered by all members of the team. Subsequently, a database was established to provide the inquiries, individual responses, and the criteria for evaluation. The presentation of Chat GPT Ellie's is uncomplicated, as each response is presented in a simple textual format.

Question: 1 of 60
Is it true that people with Alzheimer's disease are particularly prone to depression?

Date	Source	Answer	Currency	Readability	Reliability	Objectivity	Relevance
7/19/2023	Result A	Yes, people with Alzheimer's disease are at a higher risk of depression due to changes in the brain, difficulty adapting, social isolation, and other factors. It is important to recognize and treat symptoms of depression.	--Select--	Score FRS: 44.26 Score FKS: 10.07	--Select--	--Select--	--Select--
7/19/2023	Result B	Experts estimate that up to 40 percent of people with Alzheimer's disease suffer from significant depression. Identifying depression in someone with Alzheimer's can be difficult, since dementia can cause some of the same symptoms. Alzheimer's Association https://www.alz.org/caregiving/stages-behaviors/depression Depression - Alzheimer's Association	--Select--	Score FRS: 37.21 Score FKS: 11.05	--Select--	--Select--	--Select--

Similarity Between the two Results
--Select--

Previous
Next

Figure 1: Screenshot of the evaluation web application⁸⁰

In the case of Google, the advertisement results were disregarded and instead, the focus was on presenting the response box created, if available (referred to as the direct answer, as seen in Figure 1). If no direct answer was present, the top

⁷⁹ Hristidis V, Ruggiano N, Brown EL, Ganta SRR, Stewart S
ChatGPT vs Google for Queries Related to Dementia and Other Cognitive Decline: Comparison of Results J Med Internet Res 2023;25:e48966,
<https://www.jmir.org/2023/1/e48966>, accessed on 14.10.2023.

⁸⁰ Cf. Hristidis V, Ruggiano N, Brown EL, Ganta SRR, Stewart S
ChatGPT vs Google for Queries Related to Dementia and Other Cognitive Decline: Comparison of Results J Med Internet Res 2023;25:e48966,
<https://www.jmir.org/2023/1/e48966>, accessed on 14.10.2023.

organic result, consisting of the title and snippet, was displayed. The rafters were not informed of the specific platforms used because to the discrepancy in formatting (plain text and rich text). They were instructed to overlook this aspect and focus only on their evaluations.⁸¹

4.2.3 The criteria used for assessment

The rating technique employed in this study was an expansion of a preexisting rating system that had been utilized to evaluate the quality of material on websites related to dementia.⁸²

The five criteria utilized for rating purpose are as follows:

- “Currency (yes, no, not sure):”⁸³ The criterion evaluated the presence of information on the creation date of the material and its recency, namely within the last five years. The information encompassed the year of inception, the most recent revision date, and the content inside the answer itself. Responses were classified as non-current if they exhibited a deficiency in either indicating the time of content creation or if the reported date exceeded a period of five years. The evaluation of the phrase “not sure” encompassed instances in which the response contained verbiage such as “Over the course of the previous year, it has been determined that...,” whereby details on the date of inception were supplied, although we were unable to ascertain the date of publication.⁸⁴
- Clarity and comprehensibility: The provided material should possess a high degree of lucidity and be readily understandable to individuals with varying levels of literacy. In order to evaluate the readability of the responses, researchers calculated the Flesch Reading Ease Score (FRS) and the Flesch-Kincaid Grade Level Score (FKS) for each of them.⁸⁵ The Flesch Reading Score (FRS) is a numerical scale that spans from 1 to 100, with 100 being the utmost degree of reading. According to the referenced literature, publications aimed at a wide readership are typically regarded as

⁸¹ Ebd.

⁸² Cf. Yunshu Zhu, Ting Song, Ping Yu, Developing Methods to Evaluate Content Quality of Dementia Websites, <https://ebooks.iospress.nl/Download/Pdf>, accessed on 14.10.2023.

⁸³ Hristidis V, Ruggiano N, Brown EL, Ganta SRR, Stewart S ChatGPT vs Google for Queries Related to Dementia and Other Cognitive Decline: Comparison of Results J Med Internet Res 2023;25:e48966, <https://www.jmir.org/2023/1/e48966>, accessed on 14.10.2023.

⁸⁴ Cf. Yunshu Zhu, Ting Song, Ping Yu, Developing Methods to Evaluate Content Quality of Dementia Websites, <https://ebooks.iospress.nl/Download/Pdf>, accessed on 14.10.2023.

⁸⁵ Ebd..

standard when achieving a score of 60. On the other hand, a score of 70 or more is commonly perceived as indicative of ease in reading for the average adult. The FKS is determined through the conversion of the FRS into grade-level reading assessment. As an illustration, a numerical value falling within the range of 70 to 80 would correspond to an approximate reading proficiency level equivalent to that of an eight-grade student.⁸⁶

- “Reliability (reliable source, other source, no source, not sure)”⁸⁷: A answer was considered trustworthy if it included the source of the material which was recognized as a respectable source for dementia-related content, such as government websites and established organizations specializing in dementia and caring, such as the Alzheimer Association. The term “other source” pertains to instances in which an answer includes a source, such a website or organizational name, but the reliability of that source remains uncertain. The term “no source” pertained to comments that lacked any indication or details on the origin or reference of the information provided. The phrase “not sure” was employed in instances where the reply directed the reader to credible source for additional information but failed to provide any details regarding the origin of the response content. An example of such a response would be “Please reach out to the Centers for Medical and Medicaid for further details,” without specifying the source of the response content.⁸⁸
- “Objectivity (yes, no, not sure)”⁸⁹: A positive grade denoted the presence of objective knowledge devoid of any subjective emotions or personal viewpoints on the subject matter. The labels “no” and “not sure” were attributed to instances when the information originated from a commercial entity, such as Luxe Homecare, or where it was uncertain if the source belonged to a for-profit group.⁹⁰
- “Revelance (high, medium, low, not sure)”⁹¹: The rafters assessed the degree to which each response effectively addressed the specific topic

⁸⁶ Cf. Flesch R.(1948), A new readability yardstick, Journal of Applied Psychology, 32(3), 221-233, <https://psycnet.apa.org/doi/10.1037/h0057532>, accessed 16.10.2023.

⁸⁷ Hristidis V, Ruggiano N, Brown EL, Ganta SRR, Stewart S ChatGPT vs Google for Queries Related to Dementia and Other Cognitive Decline: Comparison of Results J Med Internet Res 2023;25:e48966, <https://www.jmir.org/2023/1/e48966>, accessed on 14.10.2023.

⁸⁸ Cf. Hristidis V, Ruggiano N, Brown EL, Ganta SRR, Stewart S ChatGPT vs Google for Queries Related to Dementia and Other Cognitive Decline: Comparison of Results J Med Internet Res 2023;25:e48966, <https://www.jmir.org/2023/1/e48966>, accessed on 14.10.2023..

⁸⁹ Ebd.

⁹⁰ Ebd.

⁹¹ Ebd.

posed. The raters were also requested to evaluate the resemblance between the two outcomes using the categories of “high,” “medium,” “low,” or “not sure”.⁹²

4.3 Assessment Process

The reviewer-training session was initially done with three of the writers, namely EB, NR, and SS. The authors of the study examined variations in the answers of raters in order to guarantee consistent implementation of the evaluation criteria. During the training session, the writers collectively assessed the initial 15 questions by assigning a shared rating. Subsequently, each rater separately assessed the remaining 45 questions.⁹³

4.3.1 Data Analysis

Descriptive statistics, including frequencies, were computed for all replies. The Levene test was conducted in order to evaluate the quality of variances between the Chat GPT and Google reading scores on the FRS and FKS. Subsequently, a two-tailed Welch t-test for independent samples was conducted to ascertain any disparities in the means between the two groups. The Fleiss k was calculated to assess the inter-rater agreement for the 45 questions that were scored separately.⁹⁴

4.3.2 Outcome

In order to ensure repeatability, researchers have recorded and preserved the outcomes of our inquiries conducted on Chat GPT and Google, utilizing a set of 60 predetermined questions.⁹⁵

4.4 Banknotes

In terms of the currency of information, the study revealed that among the replies generated by Chat GPT, only one response (1.7%) included the date of data

⁹² Ebd.

⁹³ Cf. C. F. Waltz et al., Measurement in Nursing and Health Research, 2010, <https://api.semanticscholar.org/CorpusID:70937398>, accessed on 16.10.2023.

⁹⁴ Cf. Hristidis V, Ruggiano N, Brown EL, Ganta SRR, Stewart S ChatGPT vs Google for Queries Related to Dementia and Other Cognitive Decline: Comparison of Results J Med Internet Res 2023;25:e48966, <https://www.jmir.org/2023/1/e48966>, accessed on 16.10.2023.

⁹⁵ Cf. Ganta SRR, Hristidis V, Brown EL, Ruggiano N. Memory query results evaluator, preview of the questions. Google Docs. 2023, <https://docs.google.com/spreadsheets/d/1fTByJ3iI0DRLJHGSEal1uXCS9jJh7P19gKc6WXxGqo/edit?usp=sharing>, accessed on 16.10.2023.

collection, while the remaining 59 responses (98.3%) did not offer any indication of the dates associated with the information presented. In the case of Google, it was found that 19 replies, accounting for 31.7% of the total, had information on the dates of data collection or source updates. Notably, these dates were seen to fall within a timeframe of the last five years. Out of the total replies, 41 (68.3%) did not contain any indication of the dates when the information was submitted, or the dates provided were found to be older than five years. When doing a comparison of current Google replies, it was observed that there was no discernible distinction between the results obtained for informational questions (n=9, 30%) and transactional queries (n=10, 33.3%).⁹⁶

Currency	Transactional questions Chat GPT, n (%)	Transactional questions Google, n (%)	Informational questions Chat GPT, n (%)	Informational questions Google, n (%)
Yes	0	10 (33.3)	0	9 (30.0)
No	30 (100.0)	20 (66.7)	30 (100.0)	21 (70.0)

Table 1: Presents the current status of replies to both informative and transactional questions.⁹⁷

4.5 Reliability of Information

In terms of dependability, Chat GPT exhibited a lack of sourcing for the information it presented in its answers. Nonetheless, a total of four replies, accounting for 6.7% of the sample, were classified as “not sure” due to the absence of source citations. Instead, these responses advised the user to seek assistance from an agency that was considered reliable. As an illustration, inquiring about the process of locating home health care services in Riverside, California. The reader was advised by Chat GPT to initiate contact with the Riverside County Office on Aging. Within the Google search engine, each result presented to the user includes the respective websites from where the information was obtained. Out of the many sources examined, a total of 36 sources,

⁹⁶ Cf. Hristidis V, Ruggiano N, Brown EL, Ganta SRR, Stewart S ChatGPT vs Google for Queries Related to Dementia and Other Cognitive Decline: Comparison of Results J Med Internet Res 2023;25:e48966, <https://www.jmir.org/2023/1/e48966>, accessed on 16.10.2023.

⁹⁷ Cf. Hristidis V, Ruggiano N, Brown EL, Ganta SRR, Stewart S ChatGPT vs Google for Queries Related to Dementia and Other Cognitive Decline: Comparison of Results J Med Internet Res 2023;25:e48966, <https://www.jmir.org/2023/1/e48966>, accessed on 16.10.2023..

accounting for 60% of the sample, were determined to be reputable. These sources, such as the Alzheimer Association and the National Council on Aging, were regarded trustworthy by the reviewers. On the other hand, the other 24 sources, or 40% of the sample, were new to the reviewers.⁹⁸

Reliability	Transactional questions Chat GPT, n (%)	Transactional questions Google, n (%)	Informational questions Chat GPT, n (%)	Informational questions Google, n (%)
No source	28 (93.3)	0	30 (100.0)	0
Not sure	2 (6.7)	0	0	0
Reliable source	0	16 (53.3)	0	25 (83.3)
Other source	0	14 (46.7)	0	5 (16.7)

Table 2: The finding of the analysis on the dependability of both informative and transactional questions.⁹⁹

In the present situation, reliability pertains to the degree to which the information presented by the source may be relied upon as being accurate and reliable. The data shown in the table indicates that there was minimal disparity seen between the replies obtained from credible sources and those obtained without any source, for both transactional and informative inquiries. In the context of informational inquiries, it was observed that the frequency of other source and not relevant replies was much greater when compared to transactional questions.¹⁰⁰

Inquiries were made on the origins of information pertaining to currency and dependability, prompting a follow-up query directed to Chat GPT (“Where did you obtain this information”). It is noteworthy that Chat GPT consistently replied by stating its utilization of a diverse array of sources to create replies, without providing any specifics.¹⁰¹

⁹⁸ Ebd.

⁹⁹ Cf. Hristidis V, Ruggiano N, Brown EL, Ganta SRR, Stewart S ChatGPT vs Google for Queries Related to Dementia and Other Cognitive Decline: Comparison of Results J Med Internet Res 2023;25:e48966, <https://www.jmir.org/2023/1/e48966>, accessed on 16.10.2023.

¹⁰⁰ Ebd.

¹⁰¹ Ebd.

4.5.1 Objectivity

All of the replies generated by Chat GPT were evaluated as exhibiting objectivity, but a majority of 49 responses (81.7%) from Google were determined to possess objectivity. Upon additional analysis, it was shown that both Chat GPT and Google exhibited comparable levels of objectivity when it came to addressing informational queries. Nevertheless, when it comes to transactional inquiries, it was determined that 2 (3.3%) of the results supplied by Google lacked objectivity. This was due to the fact that the information originated from a for-profit organization that offers services for individuals with disabilities, and there was ambiguity around whether 11 (18.3%) of the sources were also affiliated with for-profit service companies. Despite the absence of explicit promotion of the organization's services in the responses, the presence of actual or potential conflicts of interest originating from the sources has prompted inquiries regarding the degree of objectivity that can be attributed to the response. For instance, if a response to the inquiry “What criteria should I consider when selecting the most suitable home health care provider?” Was published on a website affiliated home health care services.¹⁰²

Objectivity	Transactional questions Chat GPT, n (%)	Transactional questions Google, n (%)	Informational questions Chat GPT, n (%)	Informational questions Google, n (%)
Yes	30 (100.0)	24 (82.7)	30 (100.0)	30 (100.0)
No	0	1 (3.5)	0	00
Not sure	0	4 (13.8)	0	

Table 3: Objectivity for informational and transactional questions.¹⁰³

Table 3 presents the findings pertaining to the objectivity of the two distinct types of questions employed in the present investigation. The replies to informative questions had a higher degree of congruence, whereas the responses to transactional questions displayed more variability.¹⁰⁴

¹⁰² Cf. Hristidis V, Ruggiano N, Brown EL, Ganta SRR, Stewart S
ChatGPT vs Google for Queries Related to Dementia and Other Cognitive Decline: Comparison of Results J Med Internet Res 2023;25:e48966,
<https://www.jmir.org/2023/1/e48966>, accessed on 18.10.2023.

¹⁰³ Ebd.

¹⁰⁴ Ebd.

4.5.2 Relevance

In the case of Chat GPT, the evaluators assigned a rating of high relevance to 58 replies (equivalent to 96.7%) in relation to the posed question. Additionally, two responses (equivalent to 3.3%) were deemed to have medium relevance, while none of the responses were evaluated as having low relevance. According to the evaluations provided by reviewers, a total of 36 replies (equivalent to 60% of the total) were deemed extremely relevant to the given topic. Four responses (6.7% of the total) were classified as having medium relevance, while 20 responses (33.3% of the total) were determined to have low relevance.¹⁰⁵

Relevance	Transactional questions Chat GPT, n (%)	Transactional questions Google, n (%)	Informational questions Chat GPT, n (%)	Informational questions Google, n (%)
High	30 (100.0)	22 (73.3)	28 (93.3)	23 (76.7)
Low	0	7 (23.3)	2 (6.7)	1 (3.3)
Medium	0	1 (3.3)	0	6 (20.0)

Table 4: "Relevance for informational and transactional questions".¹⁰⁶

The findings, categorized by the kind of inquiry, are shown in Table 4. The findings of the comparison indicated that the level of relevance in the replies provided by Chat GPT was comparable for both types of queries. In contrast, the responses generated by Google were more frequently evaluated as having medium relevance (n=5, 8.3%) or poor relevance (n=16, 53%) specifically for transactional inquiries. In the context of transactional inquiries, the pertinence of Google's replies was generally deemed to be lower due to its reliance on sources that mostly consisted of referral services for dementia care or the service providers themselves.¹⁰⁷

¹⁰⁵ Ebd.

¹⁰⁶ Cf. Hristidis V, Ruggiano N, Brown EL, Ganta SRR, Stewart S. ChatGPT vs Google for Queries Related to Dementia and Other Cognitive Decline: Comparison of Results J Med Internet Res 2023;25:e48966, <https://www.jmir.org/2023/1/e48966>, accessed on 18.10.2023.

¹⁰⁷ Ebd.

4.5.3 Similarity

In general, the degree of similarity between the material provided by Chat GPT and Google replies was assessed as high for 13 responses (21.7%), medium for 16 responses (26.7%), and low for 31 responses (51.6%). The findings of the study indicate that there was a higher degree of similarity in the search results for informative questions compared to transactional inquiries, as evidenced by the data presented in Table 5. The results of this study indicate that varying sorts of inquiries may necessitate distinct search techniques.¹⁰⁸

Currency	Transactional questions, n (%)	Informational questions, n (%)
High	9 (32.1)	10 (33.3)
Medium	7 (25.0)	8 (26.7)
Low	12 (42.9)	12 (40.0)

Table 5: “Similarity for informational and transactional questions”.¹⁰⁹

4.5.4 Legibility

The variability in the readability of replies from both Chat GPT and Google may be observed in Table 6.

Score and variables	Chat GPT (N=60)	Google (N=60)
FRS		
Mean (SD)	33.5 (12.46)	43.36 (19.61)
Variance	157.82	390.93
Range	4.84-57.81	4.11-91.84
FKS		
Mean (SD)	12.17 (1.94)	9.86 (3.47)
Variance	3.75	12.05

¹⁰⁸ Ebd.

¹⁰⁹ Cf. Hristidis V, Ruggiano N, Brown EL, Ganta SRR, Stewart S. ChatGPT vs Google for Queries Related to Dementia and Other Cognitive Decline: Comparison of Results J Med Internet Res 2023;25:e48966, <https://www.jmir.org/2023/1/e48966>, accessed on 18.10.2023.

Range	7.48-16.11	2.71-19.39
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Table 6: The variability in the readability of replies from Chat GPZ and Google.

In the case of Chat GPT, none of the replies exhibited a FRS equal to or more than 60, which often indicates suitability for broad reading audiences. Conversely, for Google, twelve responses, accounting for 20% of the total, had an FRS of 60 or higher. Similarly, the FKS exhibited a same pattern, with Chat GPT achieving three points (5%) and Google obtaining 21 scores (35%) at a literacy level of 8th grade or less.¹¹⁰

Nevertheless, the readability ratings of the Google replies exhibited a higher degree of fluctuation. The results of the Levene test indicated that there were statistically significant variations in the variances between the replies generated by Chat GPT and Google “for the FRS ($F_{2, 118}=11.16$, $P=.001$) and the FKS ($F_{2,118}=9.89$, $P=.002$).”¹¹¹

Consequently, a further two-tailed Welch t-test for independent samples was conducted, revealing “statistically significant differences in the FRS ($t_{100}=3.26$, $P=.001$) and the FKS ($t_{100}=4.44$, $P<.001$).”¹¹²

In general, the responses generated by Google exhibited a higher level of readability suitable for a broad range of readers. The average Flesch-Kincaid Grade Level score for Google’s responses was 9.86 (standard deviation of 3.47), indicating that, on average, the language used was at a ninth-grade reading level. In contrast, Chat GPT’s responses had an average Flesch_Kincaid Grade Level score of 12.17 (standard deviation of 1.94), corresponding to a twelfth-grade reading level.¹¹³

The results of this investigation are succinctly outlined in Table 7.

Assessment criterion	Superior performance	Comments
Currency	Google	No dates provided in Chat GPT results
Reliability	Google	No sources provided by Chat GPT

¹¹⁰ Ebd.

¹¹¹ Hristidis V, Ruggiano N, Brown EL, Ganta SRR, Stewart S ChatGPT vs Google for Queries Related to Dementia and Other Cognitive Decline: Comparison of Results J Med Internet Res 2023;25:e48966, <https://www.jmir.org/2023/1/e48966>, accessed on 18.10.2023.

¹¹² Ebd.

¹¹³ Ebd.

Objectivity	Chat GPT	Google more likely to generate responses from for-profit service agencies
Relevance	Chat GPT	Google's relevance lower for informational queries
Readability	Google	Both Chat GPT and Google requiring relatively high-grade-level preparation

Table 7: The evaluation provides a summary of the replies generated by Chat GPT and Google.¹¹⁴

4.5.5 Interrater Agreement

The Fleiss κ was computed in order to examine the level of agreement among the three raters in their evaluations of currency, dependability, objectivity, and relevance. In general, the judgments of dependability “($\kappa=0.732$, 95% CI 0.655-0.808, $P<0.001$), currency ($\kappa=0.731$, 95% CI 0.614-0.848, $P<.001$), and relevance ($\kappa=0.618$, 95% CI 0.518-0.718, $P<.001$)”¹¹⁵ demonstrated excellent levels of reliability. Furthermore, a considerable level of consensus was seen in the raters’ assessment “of objectivity ($\kappa=0.218$, 95% CI 0.115-0.320, $P<.001$).”¹¹⁶ Raters may have employed several criteria to assess impartiality, such as the source website’s affiliation with a for-profit entity, the content that produced the result, or the tone it conveyed.¹¹⁷

4.6 Examination

The main result of this study indicated that Chat GPT and Google has distinct advantages that can be beneficial for individuals experiencing Alzheimer’s disease or their caretakers, as seen in table 7. In terms of response relevance, Chat GPT

¹¹⁴ Ebd.

¹¹⁵ Hristidis V, Ruggiano N, Brown EL, Ganta SRR, Stewart S. ChatGPT vs Google for Queries Related to Dementia and Other Cognitive Decline: Comparison of Results J Med Internet Res 2023;25:e48966, <https://www.jmir.org/2023/1/e48966>, accessed on 18.10.2023.

¹¹⁶ Ebd.

¹¹⁷ Ebd.

demonstrated superior performance compared to Google. Notably, Google failed to provide an answer box for eight out of the 30 informational questions. Additionally, Chat GPT exhibited greater objectivity by generating responses that were less influenced by sources that might indicate or real monetary conflicts of interest, a tendency observed more frequently in Google's responses. Google has been gradually transitioning from keyword-based search to questions-answering (QA) systems, as seen by "the Review of Query Tools"¹¹⁸ section. Consequently, there has been a noticeable rise in the frequency of answer boxes appearing in search results for a growing range of searches. Google possesses a notable advantage in terms of currency, since it employs a continual web crawling process to gather the most up-to-date material from various online pages. However, it is worth noting that Chat GPT undergoes retraining only at regular intervals, as stated in the "Review of Query Tools section".¹¹⁹

Moreover, both platforms exhibit a deficiency in providing time stamps that indicate the currency of the shown results. Google demonstrates a higher level of reliability compared to Chat GPT due to its ability to display the sources (URLs) for each search result.¹²⁰

The importance of reliability cannot be overstated, since research has demonstrated that the prevalence for health misconceptions on the internet is substantial obstacles for individuals with physical and learning disabilities (PLWD).¹²¹

In certain instances, individuals with disabilities may choose to abstain from conducting online searches on health-related subjects due to their apprehensions over misinformation and fraudulent activities.¹²²

Individuals with disabilities have also expressed dissatisfaction with regards to accessing information on the internet pertaining to the latest study results on Alzheimer disease and associated dementias (ADRD). This discontent arises from

¹¹⁸ Ebd.

¹¹⁹ Ebd.

¹²⁰ Cf. Hristidis V, Ruggiano N, Brown EL, Ganta SRR, Stewart S. ChatGPT vs Google for Queries Related to Dementia and Other Cognitive Decline: Comparison of Results J Med Internet Res 2023;25:e48966, <https://www.jmir.org/2023/1/e48966>, accessed on 18.10.2023.

¹²¹ Cf. Emma Dixon, Jesse Anderson, Diana Blackwelder, Mary Radnofsky, and Amanda Lazar. 2022. Barriers to Online Dementia Information and Mitigation. In CHI Conference on Human Factors in Computing Systems (CHI'22), April 29–May 05, 2022, New Orleans, LA, USA. ACM, New York, NY, USA, 14 pages, <https://doi.org/10.1145/3491102.3517554>, accessed on 07.11.2023.

¹²² Cf. Emma Dixon, Anne Marie Piper, and Amanda Lazar. 2021. "Taking care of myself as long as I can": How People with Dementia Configure Self-Management Systems. In CHI Conference on Human Factors in Computing Systems (CHI '21), May 08–13, 2021, Yokohama, Japan. ACM, New York, NY, USA, 14 pages, <https://doi.org/10.1145/3411764.3445225>, accessed on 07.11.2023.

the discovery that the aforementioned conclusions were erroneous, when further research is published. This phenomenon has been documented.¹²³

The recent documentation of the influence for health deception on the internet on the Alzheimer's community has become apparent in the context of the COVID-19 pandemic. Caregivers have described instances where they declined vaccination for both themselves and their care receivers due to apprehensions stemming from online sources.¹²⁴

In addition to the currency and dependability constraints, researchers discovered a few more issues with ChatGPT.

First, it has limited support for other languages since it's model was not trained to the same degree as English.

Second, it often crashes or becomes unavailable during a discussion (as a predict that this will improve with time).

Third, it needs a monthly fee to use (after a few of free accounts were distributed), however this is gradually changing as it becomes part of the Microsoft Bing platform.

Fourth, it can't assist PLWD with any transactions, such as giving a questionnaire to assess cognitive function or locating a doctor in their region.

Fifth, when it does not completely comprehend a query, it does not ask for clarification. For example: 'Did you mean home care services?' – instead, it does it's best to reply depending on the most recent input.¹²⁵

In this research, Chat GPT comments were found to be too difficult for a broad audience to read, with readability scores averaging 12th grade compared to Google's average of 9th grade.

However, several of the Google comments were more difficult to read than what is often advised for health education materials. The raters for this research were health and social care professionals with extensive education and training.

As a result, they may have a better comprehension of the Chat GPT replies than many PLWD. Health literacy, particularly digital health literacy, has been a persistent issue in the United States, and it's a focus for Healthy People 2030,

¹²³ Cf. Emma Dixon, Jesse Anderson, Diana Blackwelder, Mary Radnofsky and Amanda Lazar. 2022. Barriers to Online Dementia Information and Mitigation. In CHI Conference on Human Factors in Computing Systems (CHI'22), April 29–May 05, 2022, New Orleans, LA, USA. ACM, New York, NY, USA, 14 pages, <https://doi.org/10.1145/3491102.3517554>, accessed on 07.11.2023.

¹²⁴ Cf. Savla J, Roberto KA, McCann BR, Blieszner R. COVID-19 Vaccination Experiences of Family Caregivers of Persons Living with Dementia in Rural Appalachia. *Journal of Applied Gerontology : the Official Journal of the Southern Gerontological Society*, <https://doi.org/10.1177/07334648221147916>, accessed on 07.11.2023.

¹²⁵ Cf. Hristidis V, Ruggiano N, Brown EL, Ganta SRR, Stewart S. ChatGPT vs Google for Queries Related to Dementia and Other Cognitive Decline: Comparison of Results *J Med Internet Res* 2023;25:e48966, <https://www.jmir.org/2023/1/e48966>, accessed on 18.10.2023.

which gives 10-year, quantifiable public health goals and tools to assist measure progress toward reaching them.¹²⁶

Furthermore, there are major gaps in health and digital health literacy,¹²⁷ exacerbating health inequities among disadvantaged PLWD. Digital literacy has been noted as a continuing difficulty for people with disabilities, as seen by COVID-19 pandemic, when many caregivers required to utilize digital platforms for health and support services.¹²⁸

The demand for technology during pandemic may have enhanced digital literacy among this demographic, the results have yet to be documented. Previous study has shown a substantial association between digital health literacy for caregivers, suggesting that strengthening digital health literacy may enhance outcomes such as self-efficacy for this group.¹²⁹

This investigation focused on the readability of the replies provided, it also revealed that many caregivers may have health and computer literacy limitations that hamper their search tactics, such as keyword use. For example, the Chat GPT replies were rated highly relevant to the questions presented in this research, other caregivers with identical information demands may not yield the same results owing to less efficient searches.¹³⁰

Chat GPT's website cites a few other restrictions, which are contributed here for completeness:

- (i) "Chat GPT sometimes writes plausible-sounding but incorrect or nonsensical answers.
- (ii) Chat GPT is sensitive to tweaks to the input phrasing or attempting the same prompt multiple times.
- (iii) The model is often excessively verbose and overuses certain phrases.
- (iv) While we've made efforts to make the model refuse inappropriate requests, it will sometimes respond to harmful instructions or exhibit biased behavior."¹³¹

¹²⁶ Cf. Jackson D, Trivedi N, Baur C. Re-prioritizing digital health and health literacy in Healthy People 2030 to Affect Health Equity, p. 1155-1162; <https://doi.org/10.1080/10410236.2020.1748828> accessed on 24.02.2024.

¹²⁷ Ebd.

¹²⁸ Cf. Chirico I, Giebel C, Lion K, Mackowiak M, Chattat R, Cations M, et al. Use of technology by people with dementia and informal carers during COVID-19: a cross-country comparison; <https://doi.org/10.1002/gps.5801> accessed on 24.02.2024.

¹²⁹ Cf. Efthymiou A, Middleton N, Charalambous A, Papastavrou E. Health literacy and eHealth literacy and their association with other caring concepts among carers of people with dementia: a descriptive correlational study, p. 1109-1119; <https://doi.org/10.1111/hsc.13341> accessed on 24.02.2024.

¹³⁰ Cf. Hristidis V, Ruggiano N, Brown EL, Ganta SRR, Stewart S

ChatGPT vs Google for Queries Related to Dementia and Other Cognitive Decline: Comparison of Results J Med Internet Res 2023;25:e48966, <https://www.jmir.org/2023/1/e48966>, accessed on 18.10.2023.

¹³¹ Introducing Chat GPT, <https://openai.com/blog/chatgpt> accessed 24.02.2024.

4.7 Moral Concerns

Several ethical difficulties arise when using Chat GPT and Google to seek for health information online.

One problem with Chat GPT is that it can provide users with erroneous or heavily biased information since it is trained on current internet material. Concerns have been expressed concerning user privacy in relation to AI browsing tools, namely around the tracking of user interactions with these technologies.¹³² Such ethical considerations have led to suggestions for a code of ethics for Chat GPT and comparable AI technologies.¹³³

Concerns about ethical concerns related to using Google for seeking health information have been brought forward as well. Google has faced criticism for sharing erroneous public health information¹³⁴, privacy issues over user data¹³⁵, and biased algorithms influencing search results. Therefore, more efforts are required to tackle ethical issues in digital health literacy overall.¹³⁶

4.8 Reciprocal Relationship

Chat GPT may provide precise health-related information, but it also has many limitations. Users may need to cross-reference replies with other sources owing to the absence of information about their origins. This might increase the need for information retrieval.¹³⁷

One possible outcome is that PLWD may choose to use Chat GPT and Google simultaneously instead of depending only on one platform over the other.¹³⁸

¹³² Cf. Ray P. ChatGPT: a comprehensive review on background, applications, key challenges, bias, ethics, limitations and future scope, p. 121-154; <https://doi.org/10.1016/j.iotcps.2023.04.003> accessed on 24.02.2024.

¹³³ Cf. Sallam M. ChatGPT utility in healthcare education, research, and practice: systematic review on the promising perspectives and valid concerns; <https://doi.org/10.3390/healthcare11060887> accessed on 24.02.2024.

¹³⁴ Cf. Morley J, Machado CCV, Burr C, Cowls J, Joshi I, Taddeo M, et al. The ethics of AI in health care: a mapping review; <https://doi.org/10.1016/j.socscimed.2020.113172> accessed on 24.02.2024.

¹³⁵ Cf. Esteve A. The business of personal data: Google, Facebook, and privacy issues in the EU and the USA, p. 36-47; <https://doi.org/10.1093/idpl/ijpw026> accessed on 24.02.2024.

¹³⁶ Cf. Lewandowski D, Sünkler S. What does Google recommend when you want to compare insurance offerings? A study investigating source distribution in Google's top search results, p.310-324, <https://doi.org/10.1108/AJIM-07-2018-0172> accessed on 24.02.2024.

¹³⁷ Cf. Lee P, Bubeck S, Petro J. Benefits, limits, and risks of GPT-4 as an AI chatbot for medicine, p. 1233-1239; https://www.nejm.org/doi/10.1056/NEJMSr2214184#article_citing_articles accessed on 25.02.2024.

¹³⁸ Ebd.

4.9.1 Forecast towards the Future

In the near future, chatbot systems like Chat GPT and online search engines like Google are anticipated to gradually merge. Chatbots will provide prompt responses, provide sources such as web page URLs, and give multimedia interfaces like photos and videos. Web search engines will transition from displaying lists of sites to providing answers as search results.

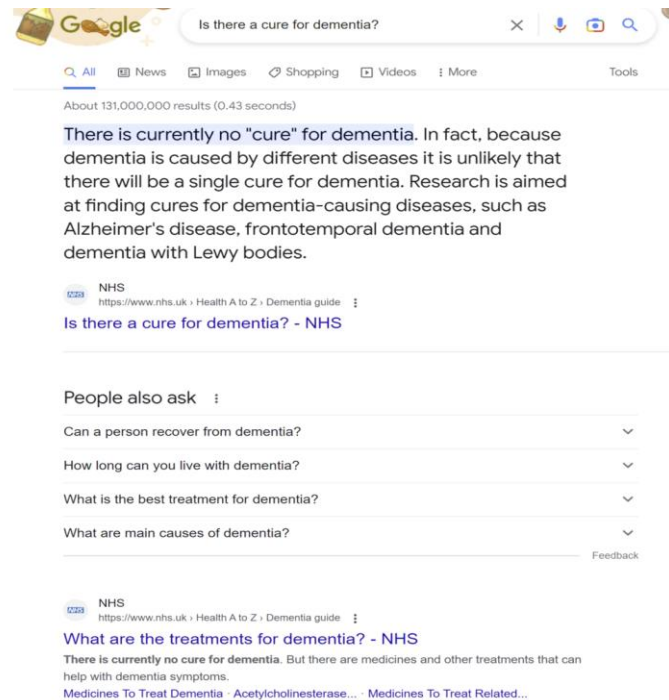


Figure 2: Is there a cure for dementia?¹³⁹

The results page will start to prioritize the top answer (answer box) more, making it more accurate, while reducing the importance of the following list of websites, Web search engines will improve in using the search or conventional history of the user.¹⁴⁰

4.9 Results

This research examined the efficacy of Google and Chat GPT in responding to inquiries on dementia or cognitive decline. Three professionals evaluated the results of 60 informative and transactional searches based on various criteria. Both Google and Chat GPT exhibit advantages and disadvantages.

¹³⁹ Cf. Google search engine accessed on 25.02.2024.

¹⁴⁰ Cf. Hristidis V, Ruggiano N, Brown EL, Ganta SRR, Stewart S. ChatGPT vs Google for Queries Related to Dementia and Other Cognitive Decline: Comparison of Results J Med Internet Res 2023;25:e48966, <https://www.jmir.org/2023/1/e48966>, accessed on 18.10.2023.

Google often includes the source and date of the answer, but Chat GPT demonstrates superior response accuracy and neutrality. Their amalgamation may provide superior outcomes.

Additional research and innovative technologies are required to enhance the language comprehension and accuracy of Chat GPT, integrating it with the extensive coverage and up-to-date information of Google.¹⁴¹

5. Case Studies: Successful implementation of Chat GPT in chronic disease management

5.1 Case Study 1: Using Chat GPT for personalized diabetes management

Managing chronic illness like diabetes presents significant problems for both patients and healthcare professionals. Due to the growing number of chronic illness, there is a greater need for creative solutions that may improve the quality of treatment and enable patients to manage their health independently. This case study examines the use of Chat GPT in personalized diabetes treatment, demonstrating its ability to provide customized advice, track symptoms, and offer lifestyle suggestions.

5.1.1 Diabetes

Diabetes is a persistent condition marked by elevated amounts of glucose in the bloodstream.

Typ 1 diabetes results from pancreas ceasing to create insulin, whereas Typ 2 diabetes arises when the body's cells do not react well to the insulin produced. Individuals with diabetes have an increased likelihood of acquiring cardiovascular conditions including heart attack and stroke if the condition is undetected or inadequately managed. There are also at increased risk of vision impairment, amputation of the foot and leg, and kidney failure.¹⁴²

In 2017, there were around 32.7 million individuals with diabetes in the European Union (EU), an increase from around 18.2 million persons in 2000.

¹⁴¹ Ebd.

¹⁴² Cf. OECD/EU (2018), Health at a Glance: Europe 2018: State of Health in the EU Cycle, OECD Publishing, Paris. https://doi.org/10.1787/health_glance_eur-2018-en accessed on 25.02.2024.

Furthermore, an estimated 12.8 million individuals had undiagnosed diabetes in 2017.¹⁴³

The prevalence of diagnosed diabetes in males has significantly risen since 2000, more than doubling from around 8 million in 2000 to 17.1 million in 2017.¹⁴⁴

The number of women diagnosed with diabetes increased significantly from 10.3 million in 2000 to 15.6 million in 2017, representing a growth of more than 50%.¹⁴⁵

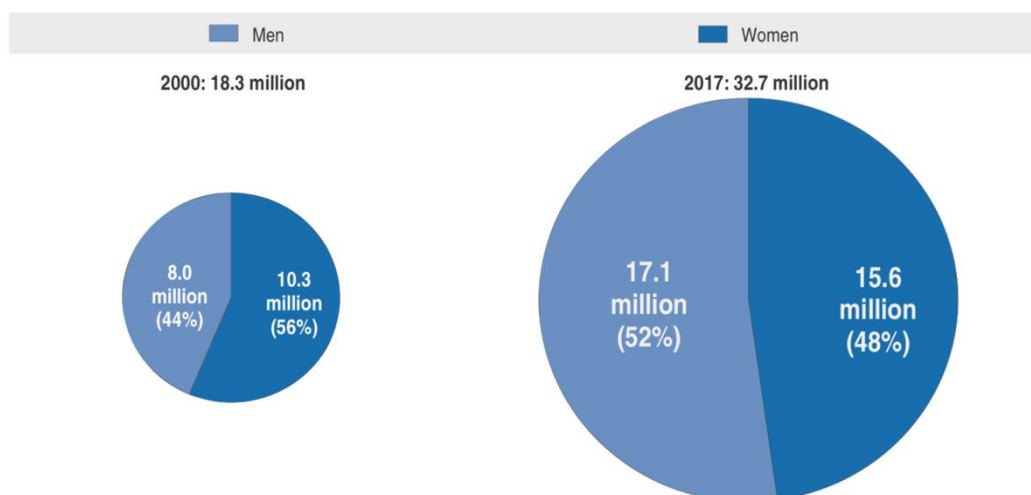


Figure 3: Diabetes prevalence in the EU28 between 2000 and 2017¹⁴⁶

Diabetes is more prevalent in elderly individuals: 19.3 million persons between the ages of 60-79 have diabetes in EU nations, compared to 11.7 million individuals aged 40-59 and just 1.8 million aged 20-39.

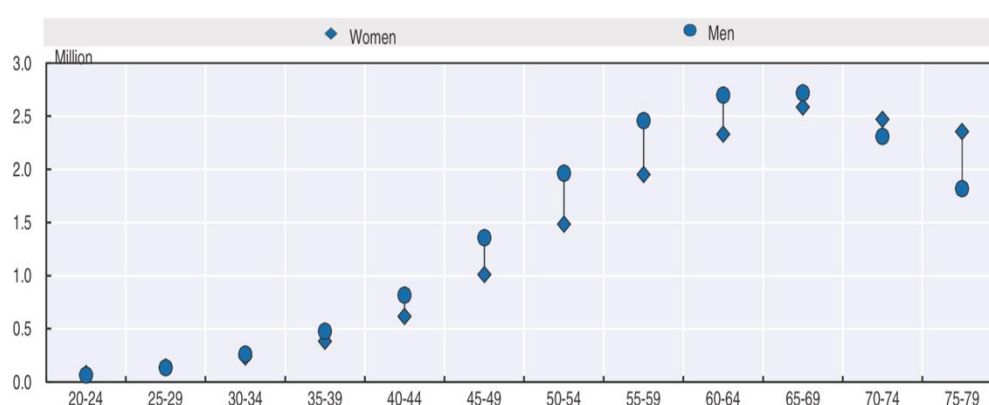


Figure 4: Diabetes prevalence in EU28 in 2017, categorized by gender and age group¹⁴⁷

¹⁴³ Ebd.

¹⁴⁴ Cf. OECD/EU (2018), Health at a Glance: Europe 2018: State of Health in the EU Cycle, OECD Publishing, Paris. https://doi.org/10.1787/health_glance_eur-2018-en accessed on 25.02.2024.

¹⁴⁵ Cf. OECD/EU (2018), Health at a Glance: Europe 2018: State of Health in the EU Cycle, OECD Publishing, Paris. https://doi.org/10.1787/health_glance_eur-2018-en accessed on 25.02.2024.

¹⁴⁶ Cf. IDF Atlas, 8th Edition, 2017 and OECD estimates; <http://dx.doi.org/10.1787/888933834832> accessed on 25.02.2024.

¹⁴⁷ Cf. IDF Atlas, 8th Edition, 2017; <http://dx.doi.org/10.1787/888933834851> accessed on 25.02.2024.

Men have a higher prevalence of diabetes than women in middle-age (40-59 years old), while women surpass men in diabetes cases beyond age 70 due to their longer lifespan.¹⁴⁸

In 2017, the average age-standardized prevalence rate of diabetes among adults in EU nations was 6%. Rates ranged from 9% or higher in Portugal, Romania, and Malta to 4% or lower in Ireland, Lithuania, and Estonia.¹⁴⁹

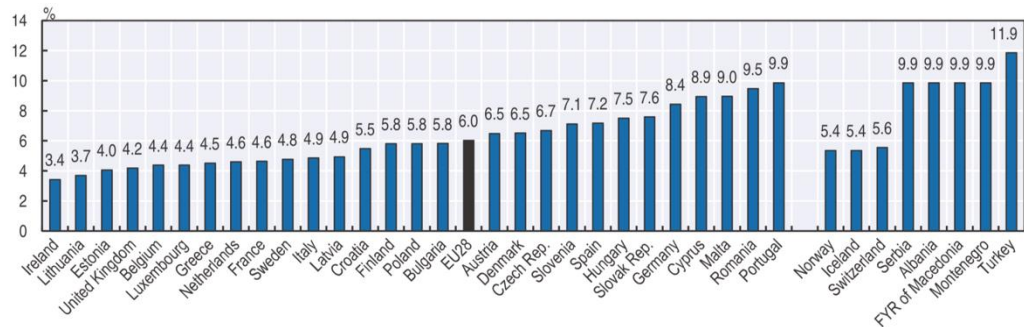


Figure 5: Diabetes prevalence among adults in 2017¹⁵⁰

Diabetes prevalence rates have remained steady in some European nations, particularly in Nordic countries, while seeing a minor increase in Southern Europe and Central and Eastern Europe. The increase in obesity and lack of physical exercise, together with the aging population, are contributing factors to these rising trends (NCD Risk Factor Collaboration, 2016).¹⁵¹

According to self-reported data from the 2014 European Health Interview Survey's second wave, adults with the lowest education level are over twice as likely to report having diabetes compared to those with the highest education level, on average in EU countries.¹⁵²

This might be attributed to a greater percentage of individuals with lower levels of education in older age groups and the correlation between age and an increased risk of diabetes. Individuals with lesser levels of education have a greater frequency of significant risk factors for diabetes, such as obesity.¹⁵³

Diabetes imposes a significant economic cost. In 2017, the European Union spent over 150 billion Euro on healthcare for diabetes treatment and complication

¹⁴⁸ Cf. OECD/EU (2018), Health at a Glance: Europe 2018: State of Health in the EU Cycle, OECD Publishing, Paris. https://doi.org/10.1787/health_glance_eur-2018-en accessed on 25.02.2024.

¹⁴⁹ Cf. OECD/EU (2018), Health at a Glance: Europe 2018: State of Health in the EU Cycle, OECD Publishing, Paris. https://doi.org/10.1787/health_glance_eur-2018-en accessed on 25.02.2024.

¹⁵⁰ Cf. IDF Atlas, 8th Edition; <http://dx.doi.org/10.1787/888933834870> accessed 25.02.2024.

¹⁵¹ Cf. OECD/EU (2018), Health at a Glance: Europe 2018: State of Health in the EU Cycle, OECD Publishing, Paris. https://doi.org/10.1787/health_glance_eur-2018-en accessed on 25.02.2024.

¹⁵² Ebd.

¹⁵³ Ebd.

prevention. The average annual cost per diabetic adult was estimated at around 4.600,00 Euro (IDF, 2017).¹⁵⁴

Type 2 diabetes is highly avoidable. Several risk factors, including as being overweight or obese, poor diet, and lack of physical exercise, may be changed.

The occurrence of overweight and obesity is on the rise in many nations.

These emphasize the need of implementing successful preventative measures.¹⁵⁵

5.1.2 Application of Chat GPT in Diabetes Management

Chat GPT may function as a virtual assistant, interacting with patients to comprehend their individual situations and provide customized guidance.¹⁵⁶

Chat GPT use machine learning algorithms to analyze extensive data and customize replies to meet individual requirements, guaranteeing precise and relevant information for patients.¹⁵⁷

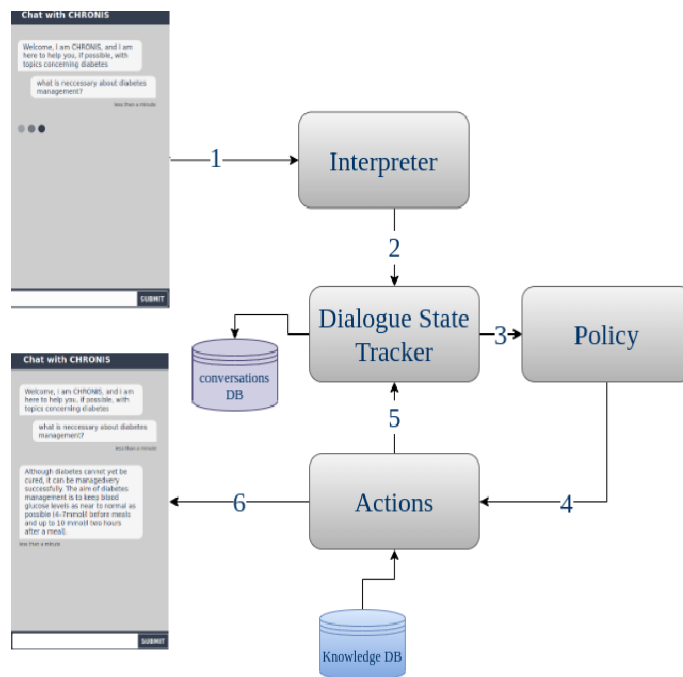


Figure 6: EVA's high level architecture.¹⁵⁸

¹⁵⁴ Ebd.

¹⁵⁵ Cf. OECD/EU (2018), Health at a Glance: Europe 2018: State of Health in the EU Cycle, OECD Publishing, Paris. https://doi.org/10.1787/health_glance_eur-2018-en accessed on 25.02.2024.

¹⁵⁶ Cf. Ramchandani N. Virtual Coaching to Enhance Diabetes Care, Diabetes Technol Ther. 21.06.2019 (S2); p.248-251; accessed on 18.03.2024.

¹⁵⁷ Ebd.

¹⁵⁸ Cf. Anastasiadou, Magdalini et al. "A prototype educational virtual assistant for diabetes management." 2020 IEEE 20th International Conference on Bioinformatics and Bioengineering (BIBE) (2020): 999-1004, <https://doi.org/10.1109/BIBE50027.2020.00169>, accessed on 25.03.2024.

5.1.3 Monitoring Symptoms

Chat GPT may function as a digital companion, inquiring about patients' problems, monitoring their blood sugar levels, and offering immediate feedback. By engaging in ongoing conversation, patients may acquire valuable knowledge about how their lifestyle choices affect their illness, empowering them to make well-informed decisions.¹⁵⁹

5.1.4 Lifestyle Recommendations

Effectively managing diabetes often necessitates implementing substantial lifestyle modifications, such as embracing a nutritious diet and intergrating consistent physical activity. Chat GPT may provide tailored lifestyle advice by taking into account an individual's preferences, medical history, and current health condition. Chat GPT offers practical ideas and motivational assistance to encourage people in taking control of their diabetes by comprehending the distinct obstacles presented by each patient.¹⁶⁰

5.1.5 Conclusion

In the pursuit of improving chronic disease management, Chat GPT proves to be a potent instrument for providing tailored diabetic treatment in the healthcare industry.¹⁶¹ Chat GPT has the potential to transform the management of diabetes by giving personalized guidance, monitoring symptoms, and offering lifestyle suggestions. Through further study and development, the healthcare industry can fully use Chat GPT to enhance results and empower people in their pursuit of optimum health.¹⁶²

5.2 Case study 2: Chat GPT as a Virtual Assistant for Asthma Patients

Recently, the healthcare industry has seen substantial progress in the field of AI, namely in the development of chatbots. Intelligent virtual assistants have the

¹⁵⁹ Cf. Gong, Enying et al. "My Diabetes Coach, a Mobile App-Based Interactive Conversational Agent to Support Type 2 Diabetes Self-Management: Randomized Effectiveness-Implementation Trial." *Journal of Medical Internet Research* 22 (2020): n. pag., <https://doi.org/10.2196/20322>, accessed on 18.03.2024.

¹⁶⁰ Cf. Rose, Vanessa et al. "A better model of diabetes self-management? Interactions between GP communication and patient self-efficacy in self-monitoring of blood glucose." *Patient education and counseling* vol. 77.2 (2009): 260-5, <https://doi.org/10.1016/j.pec.2009.03.026>, accessed on 18.03.2024.

¹⁶¹ Cf. Sharma, Samridhi et al. "A Critical Review of ChatGPT as a Potential Substitute for Diabetes Educators." *Cureus* 15 (2023), accessed on 25.03.2024.

¹⁶² Cf. Anastasiadou, Magdalini et al. "A prototype educational virtual assistant for diabetes management." *2020 IEEE 20th International Conference on Bioinformatics and Bioengineering (BIBE)* (2020): 999-1004, <https://doi.org/10.1109/BIBE50027.2020.00169>, accessed on 25.03.2024.

capacity to transform chronic illness management via the provision of individualized advice, monitoring of symptoms, and offering lifestyle suggestions.¹⁶³

This case study examines the use of Chat GPT as a virtual assistant for persons with asthma, providing insight into how this technology may improve the quality of life for those with long-term health concerns.

5.2.1 Asthma

A person has airway inflammation, narrowing, and swelling, resulting in the production of excess mucus, which leads to breathing difficulties.

Asthma may range from being mild to significantly impacting everyday activities.¹⁶⁴ Occasionally, it may result in a potentially fatal assault.

Asthma is a significant public health challenge for both industrialized and developing nations. Globally, around 300 million individuals are afflicted with bronchial asthma.¹⁶⁵

The term asthma originates from the Greek language and signifies panting or labored breathing. Asthma is a medical disorder that is characterized by sudden and recurring episodes of wheezing and difficulty breathing, especially during exhalation.¹⁶⁶

5.2.2 Causes

Based on its etiology, bronchial asthma is classified into the following groups:

1. Allergic Asthma – This kind of asthma often begins during infancy and is frequently preceded by eczema. However a significant proportion of individuals under the age of 35 who are developing asthma also belong to this group. Genetic factors are also very influential in this matter. This particular kind of asthma is characterized by the allergen triggering an overproduction of immunoglobulins, namely IgE.¹⁶⁷

¹⁶³ Cf. Kowatsch, Tobias et al. "A digital assistant for healthcare providers targeting 10 to 15-year-old patients with asthma and their family: results from a pilot study." (2019), <https://doi.org/10.3929/ETHZ-B-000368540>, accessed on 25.03.2024.

¹⁶⁴ Cf. Padem, Nurciek and Carol Anne Saltoun. "Classification of asthma." *Allergy and asthma proceedings* 40 6 (2019): 385-388, <https://doi.org/10.2500/aap.2019.40.4253>, accessed on 03.04.2024.

¹⁶⁵ Ebd.

¹⁶⁶ Cf. Willson, Graham Fairlie. "Asthma in Schoolchildren." *The Journal of the Royal Society for the Promotion of Health* 78 (1958): 274 – 282, <https://doi.org/10.1177/146642405807800330>, accessed on 03.04.2024.

¹⁶⁷ Cf. Dahat, Archana B. "An overview of asthma and its miasmatic classification." *International Journal of Homoeopathic Sciences* (2021), <https://doi.org/10.33545/26164485.2021.V5.IIE.331>, accessed on 03.04.2024.

2. Infective or Intrinsic Asthma – This kind of asthma is neither inherited or caused by allergies, but it may be triggered by, or at least linked to, upper respiratory tract or bronchial infections, often of viral origin.¹⁶⁸
3. Emotional Asthma – also known as psychological asthma, is a condition where psychological elements such as worry and emotional stress are believed to be the primary cause of certain asthmatic episodes. However, it is still uncertain if these variables might be the main cause or whether they just contribute to triggering the attacks.¹⁶⁹
4. Occupational asthma – may develop in certain sectors where workers are exposed to substances such as metallic dusts, particularly platinum salts, biological detergents, toluene diisocyanate, polyurethane, wheat and dust from grains.¹⁷⁰

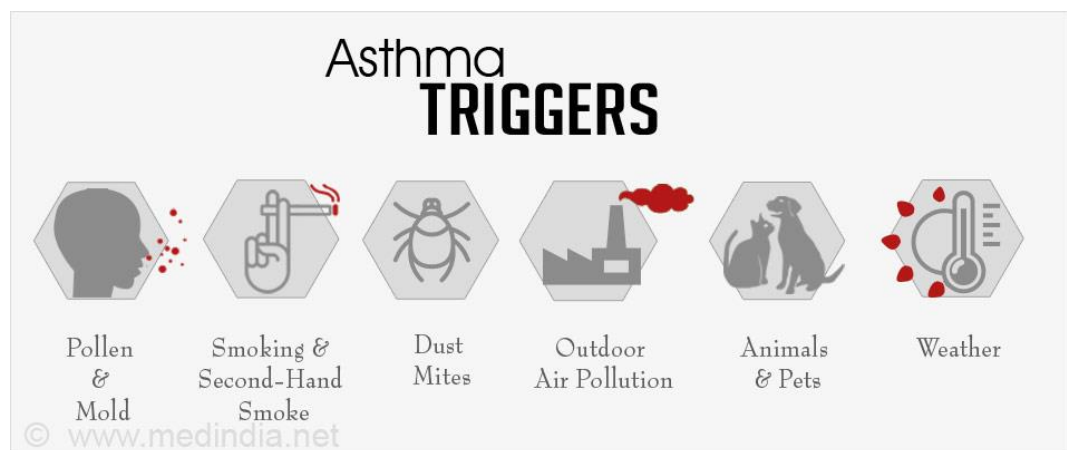


Figure 7: Factors that Trigger an attack of Asthma.¹⁷¹

5.2.3 Application of Chat GPT as a Virtual Assistant for Asthma Patients

Chat GPT, with OpenAI's cutting-edge language model, provides a potential resolution to these difficulties. Chat GPT utilizes natural language processing and machine learning algorithms to interact with asthma sufferers, offering tailored assistance and advice.¹⁷²

¹⁶⁸ Ebd.

¹⁶⁹ Ebd.

¹⁷⁰ Cf. Dahat, Archana B. "An overview of asthma and its miasmatic classification." *International Journal of Homoeopathic Sciences* (2021), <https://doi.org/10.33545/26164485.2021.V5.I1E.331>, accessed on 03.04.2024.

¹⁷¹ Cf. Dr. Reeja Tharu, M.Phil., Ph.D "Factors that trigger Asthma." The Medindia Medical Review Team (2023), <https://www.medindia.net/health/conditions/asthma-factors.htm>, accessed on 09.04.2024.

¹⁷² Cf. Kowatsch, Tobias et al. "A digital assistant for healthcare providers targeting 10 to 15-year-old patients with asthma and their family: results from a pilot study." (2019), <https://doi.org/10.3929/ETHZ-B-000368540>, accessed on 09.04.2024.

5.2.4 Monitoring Systems

An essential aspect of Chat GPT is its capability to monitor symptoms. Patients have the option to express their symptoms either via written text or spoken words, and the virtual assistant may evaluate the provided immediate response. Chat GPT has the ability to evaluate the seriousness of symptoms, provide prompt suggestions for adjusting medicine or taking emergency measures, and even advise on whether to seek medical assistance.¹⁷³

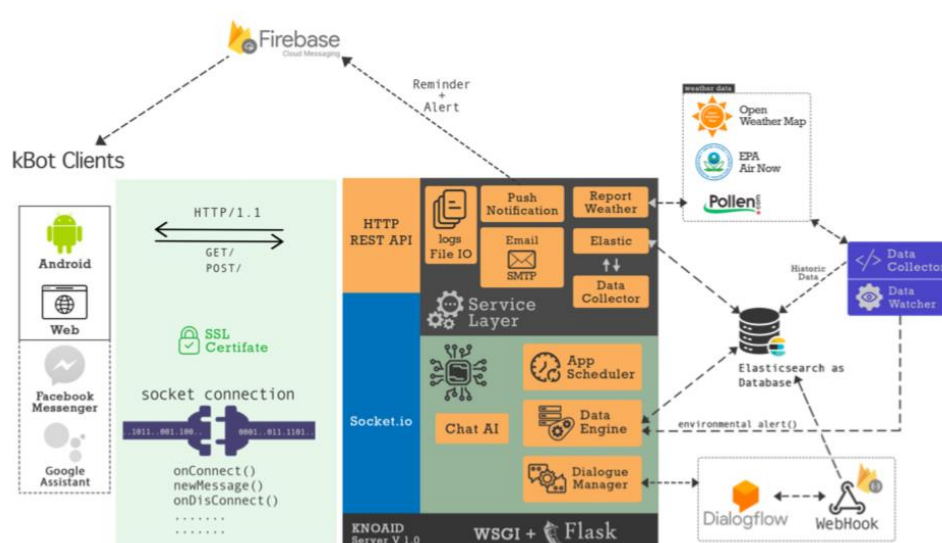


Figure 8: “System architecture and workflow of kBot.”¹⁷⁴

The pipeline starts with the patient engaging in conversation with the front-end Android (client), while data is sent to the backend server responsible for managing the conversation. A client establishes communication with the server by using an SSL encrypted socket layer. The patient data obtained from the chat is saved in Elasticsearch which is a NoSQL database, in JSON format. The server’s service layer provides clients with various services such as notification, email, data storage, and weather reporting.¹⁷⁵

Kbot archives individual chat logs with each user in a file for manual examination. It utilizes a client-server architecture (Figure 8) in which the client is

¹⁷³ Cf. Kadariya, Dipesh. “kBot: Knowledge-Enabled Personalized Chatbot for Self-Management of Asthma in Pediatric Population.” (2019). https://corescholar.libraries.wright.edu/cgi/viewcontent.cgi?article=3249&context=etd_all, accessed on 09.04.2024.

¹⁷⁴ Cf. Kadariya, Dipesh. “kBot: Knowledge-Enabled Personalized Chatbot for Self-Management of Asthma in Pediatric Population.” (2019). https://corescholar.libraries.wright.edu/cgi/viewcontent.cgi?article=3249&context=etd_all, accessed on 09.04.2024.

¹⁷⁵ Ebd.

a lightweight chat interface, and the backend server is a separate web application hosted in the cloud. Users may engage with kBot via the client application using both text and voice communication modalities.¹⁷⁶

During the initial establishment of a user profile, a unique CHAT ID is provided to each user. This ID serves as the main identifier for the user. The ID is created in the client application using the MD5 hashing technique. This process takes a text of any length as input and encodes it into a 128-bit fingerprint, which is the result of the hash. The client utilizes it to verify the communication request with the server and sustain the user session.¹⁷⁷

The client engages in bi-daily communication with patients and records all patient-specific conversation history on the server. JSON 1, a compact format for exchanging data, is used for communication between clients and servers. The relevant data is taken from the chat logs stored on the server. Simultaneously, the server consistently monitors and gathers environmental data, at variously intervals, using a third-party weather API for the zip codes that are distinct to each patient.¹⁷⁸

5.2.5 Lifestyle Recommendations

Chat GPT may serve as a complete tool for those with asthma. The system may provide up-to-date information on the most recent treatment protocols, address often asked issues, and give personalized lifestyle suggestions based on each individual's specific situation. As an instance, the virtual assistant has the capability to recommend specific activities that may enhance lung function or provide advice on how to create a home environment that is conducive to managing asthma.¹⁷⁹

Chat GPT is quite proficient in delivering tailored guidance. The system may consider the medical history, current medicines, and lifestyle characteristics of each patient to provide customized suggestions. Through the process of continual learning from data and user interactions, Chat GPT may enhance and improve its replies, resulting in more precise and relevant advice for patients.¹⁸⁰

¹⁷⁶ Ebd.

¹⁷⁷ Cf. Kadariya, Dipesh. "kBot: Knowledge-Enabled Personalized Chatbot for Self-Management of Asthma in Pediatric Population." (2019). https://corescholar.libraries.wright.edu/cgi/viewcontent.cgi?article=3249&context=etd_all, accessed on 09.04.2024.

¹⁷⁸ Ebd.

¹⁷⁹ Cf. Calvo, Rafael A et al. "Assessing the Feasibility of a Text-Based Conversational Agent for Asthma Support: Protocol for a Mixed Methods Observational Study." JMIR Research Protocols 12 (2022):n.pag., <https://doi.org/10.2196/42965>, accessed on 09.04.2024.

¹⁸⁰ Ebd.

5.2.6 Conclusion

Ultimately, using Chat GPT as a virtual assistant for individuals with asthma has significant promise in the treatment of long-term health conditions. Through the provision of tailored guidance, tracking of symptoms, and offering suggestions for lifestyle adjustments, this technology has the potential to enable persons with asthma to assume command of their illness and enhance their general state of health. With the adoption of AI-driven solutions in the healthcare industry, the outlook for chronic illness management seems more promising than ever.¹⁸¹

5.3 Case study 3: Enhancing Chronic Pain Management with Chat GPT

Chronic pain has a profound influence on the lives and well-being of millions of people worldwide. The healthcare industry has been actively seeking novel ways to assist persons with chronic illnesses, and the emergence of Chat GPT has shown encouraging outcomes in this aspect. This case study seeks to investigate the potential of Chat GPT in improving chronic pain treatment by the provision of individualized guidance, monitoring of symptoms, and offering lifestyle suggestions.

5.3.1 Application

A major obstacle in the treatment of chronic pain is the absence of customized advice that is specifically designed to meet the distinct requirements of each individual. Chat GPT can resolve this problem by using its natural language processing skills to participate in meaningful dialogues with patients. Chat GPT may provide tailored guidance for pain management by comprehending unique pain experiences. This may include recommending certain exercises, relaxation methods, or adjustments to one's lifestyle.¹⁸²

¹⁸¹ Cf. Taylor, Michael J et al. "Virtual worlds to support patient group communication? A questionnaire study investigating potential for virtual world focus group use by respiratory patients." *British journal of educational technology* : journal of the Council for Educational Technology vol. 48,2 (2017); <https://doi.org/10.1111/bjet.12378>, accessed on 09.04.2024.

¹⁸² Cf. Hauser-Ulrich, Sandra et al. "A Smartphone-Based Health Care Chatbot to Promote Self-Management of Chronic Pain (SELMA): Pilot Randomized Controlled Trial." *JMIR mHealth and uHealth* 8 (2019): n. pag., <https://doi.org/10.2196/preprints.15806>, accessed on 09.04.2024.

5.3.2 Monitoring Symptoms

Chat GPT might have a significant impact in overseeing symptoms and assessing the course of therapy. Patients may often engage in communication with Chat GPT to report their pain levels, identify triggers, and document any changes in symptoms. The real-time data may be very important for healthcare practitioners as it allows them to evaluate the effectiveness of treatments, make any required modifications, and provide prompt interventions. In addition, Chat GPT has the ability to educate patients on the significance of self-monitoring and encourage them to actively engage in their pain management process.¹⁸³

Medical diagnostics is the procedure of discovering and ascertaining the underlying cause of a patient's health condition symptoms. The diagnosis method often involves a blend of evaluations of medical history, physical exams, laboratory tests, imaging investigations, and genetic testing. Medical diagnostics are used to identify various health issues, including infectious infections, hereditary disorders, chronic illness, and acute accidents. The primary goal of medical diagnostics is to precisely determine a diagnosis for patient's condition to proper therapy. The Functional Medicine Master Health Survey (FM MHS) is an extensive health questionnaire and through blood tests used by Functional Medicine Practitioners (FMPs) to evaluate a patient's health condition and detect any underlying health issues.¹⁸⁴

5.3.3 Lifestyle recommendations

This tool assists FMPs in comprehending a patient's medical background, pattern of symptoms, and lifestyle behaviors in order to develop a customized treatment plan. The FM MHS includes inquiries on several determinants that might impact health, including as a diet, sleep, physical activity, stress, emotional welfare, environmental exposures, and a comprehensive medical history. Additionally, it encompasses inquiries on certain symptoms and ailments, such as gastrointestinal problems, endocrine disorders, and heart disease. The survey is usually conducted with patients prior to their first appointment with a Family Medicine Practitioner

¹⁸³ CF. Hauser-Ulrich, Sandra et al. "A Smartphone-Based Health Care Chatbot to Promote Self-Management of Chronic Pain (SELMA): Pilot Randomized Controlled Trial." *JMIR mHealth and uHealth* 8 (2019): n. pag., <https://doi.org/10.2196/preprints.15806>, accessed on 09.04.2024.

¹⁸⁴ Cf. Kumaresan, A., Uden, L., Ashraf, S. (2023). Potential Role of ChatGPT in Healthcare in the Prevention and Management of Non-communicable Diseases. In: Uden, L., Ting, I.H. (eds) *Knowledge Management in Organisations*. KMO 2023. Communications in Computer and Information Science, vol 1825. Springer, Cham. https://doi-org.printkr.hs-niederrhein.de/2443/10.1007/978-3-031-34045-1_34, accessed on 10.04.2024.

(FMP). The practitioners evaluate the patient's replies to detect any underlying imbalances and creates a personalized treatment plan that targets the root cause of the patient's health issues. The FM MHS is an invaluable resource of FMPs since it offers a thorough assessment of patient's health condition and aids in identifying areas that may need further testing or action. Additionally, it aids healthcare professionals in monitoring advancements over a period and modifying treatment strategies to enhance patient results. Therefore, enhancing the GPT's capacity to comprehend the user's reaction to the FM MHS and evaluate the outcome would allow FMPs to analyze data effectively.¹⁸⁵

5.3.4 Conclusion

Ultimately, Chat GPT have significant potential to improve the treatment of chronic pain. Through the provision of tailored guidance, the monitoring symptoms, and the offering of suggestions for lifestyle adjustments, persons with chronic diseases may be empowered to actively control their pain and enhance their overall well-being. The healthcare industry may use this technology to close the divide between patients and healthcare professionals, guaranteeing more efficient and tailored treatment of chronic disease. Through more investigation and advancement, Chat GPT has the potential to completely transform the approach to managing chronic pain, providing a fresh channel of assistance and treatment for persons with long-term health disorders.¹⁸⁶

6. Ethical Considerations and Privacy Concerns in Chat GPT Usage

Firstly, it is crucial in order to establish precise definitions for certain fundamental words that are essential to the discourse on security and privacy. The words responsible AI, ethical AI, trustworthy AI, and AI alignment are used to describe different aspects of artificial intelligence. Responsible AI refers to the development and implementation of AI systems in a way that upholds ethical standards, ensures transparency, and establishes accountability.¹⁸⁷

¹⁸⁵ Cf. Kumaresan, A., Uden, L., Ashraf, S. (2023). Potential Role of ChatGPT in Healthcare in the Prevention and Management of Non-communicable Diseases. In: Uden, L., Ting, I.H. (eds) Knowledge Management in Organisations. KMO 2023. Communications in Computer and Information Science, vol 1825. Springer, Cham. https://doi-org.printkr.hs-niederrhein.de:2443/10.1007/978-3-031-34045-1_34, accessed on 10.04.2024.

¹⁸⁶ Ebd.

¹⁸⁷ Cf. Huang, K., Duan, Y. (2023). ChatGPT in Nutrition Science. In: Huang, K., Wang, Y., Zhu, F., Chen, X., Xing, C. (eds) Beyond AI. Future of Business and Finance. Springer, Cham. https://doi-org.printkr.hs-niederrhein.de:2443/10.1007/978-3-031-45282-6_6, accessed on 10.04.2024.



Figure 9: “The mind map of Chapter 11.”¹⁸⁸

The approach recognizes the influence of AI on society and aims to reduce adverse effects by emphasizing factors like as equity, explainability, and resilience. Ethical AI, which is closely associated with responsible AI, is the guiding notion that AI systems should adhere to universal values and standards. It includes elements such as equity, openness, and honoring confidentiality.

Ethical AI encompasses not just the technological aspects, but also considers its ramifications and the manner in which it is used.

Trustworthy AI refers to artificial intelligence system that functions in a manner that inspires confidence and trust among people. The concept encompasses elements like as dependability, resilience, security, confidentiality, and openness. A reliable AI regularly demonstrates anticipated behavior, ensures privacy protection, and makes judgments that are both explainable and verifiable.¹⁸⁹

AI alignment, however, focuses on guaranteeing that the objectives of AI systems are in accordance with human values. This addresses the task of creating AI systems that will exhibit desirable behavior, even as they grow more independent and sophisticated. Although these notions have a shared objective of assuring

¹⁸⁸ Huang, K., Duan, Y. (2023). ChatGPT in Nutrition Science. In: Huang, K., Wang, Y., Zhu, F., Chen, X., Xing, C. (eds) Beyond AI. Future of Business and Finance. Springer, Cham. https://doi-org.printkr.hs-niederrhein.de/2443/10.1007/978-3-031-45282-6_6, accessed on 10.04.2024.

¹⁸⁹ Ebd.

responsible and ethical development and deployment of AI, they have different areas of emphasis. Ethical AI prioritizes values, Responsible AI focuses on reducing adverse effects, Trustworthy AI revolves on traits that foster trust, and AI alignment primarily concerns the compatibility of goals with human values.¹⁹⁰

6.1 Ensuring Patient Privacy and Data Security

Ensuring patient privacy and data security is of paramount significance in the constantly changing field of healthcare technology. As the healthcare industry investigates the possibilities of using Chat GPT to improve chronic illness management, it is essential to acknowledge and resolve the problems around the privacy and security of patient data.¹⁹¹

The promise of Chat GPT to transform chronic illness treatment lies in its capacity to provide tailored advice, monitor symptoms, and deliver lifestyle suggestions. Nevertheless, healthcare institutions must implement proactive strategies to ensure the security of patient privacy and secure sensitive data.¹⁹²

The previous part presented a comprehensive summary of the security considerations associated with Chat GPT. This section, will thoroughly explore specific security vulnerabilities, acknowledging that these examples may not cover every possible scenario. It is crucial to acknowledge that the mentioned risks are only a limited selection, since the ever-changing realm of AI-powered technology consistently introduces novel types of assaults and weakness. To effectively navigate the changing landscape of AI security, it is crucial to comprehend the various cyberattacks that can occur, including phishing attacks, the creation of fraudulent websites, impersonation, manipulation of user trust and deception, deep fake attacks, model poison attacks, model API attacks, and prompt injection attacks. Overlooking these dangers in Figure 10 and the hazards they entail, while acknowledging the need of being watchful and adaptable as Chat GPT grows more prevalent in both corporate and consumer sectors.¹⁹³

¹⁹⁰ Cf. Huang, K., Duan, Y. (2023). ChatGPT in Nutrition Science. In: Huang, K., Wang, Y., Zhu, F., Chen, X., Xing, C. (eds) Beyond AI. Future of Business and Finance. Springer, Cham. https://doi-org.printkr.hs-niederrhein.de:2443/10.1007/978-3-031-45282-6_6, accessed on 10.04.2024.

¹⁹¹ Cf. Kumaresan, A., Uden, L., Ashraf, S. (2023). Potential Role of ChatGPT in Healthcare in the Prevention and Management of Non-communicable Diseases. In: Uden, L., Ting, I.H. (eds) Knowledge Management in Organisations. KMO 2023. Communications in Computer and Information Science, vol 1825. Springer, Cham. https://doi-org.printkr.hs-niederrhein.de:2443/10.1007/978-3-031-34045-1_34, accessed on 10.04.2024.

¹⁹² Ebd.

¹⁹³ Cf. Huang, K., Duan, Y. (2023). ChatGPT in Nutrition Science. In: Huang, K., Wang, Y., Zhu, F., Chen, X., Xing, C. (eds) Beyond AI. Future of Business and Finance. Springer, Cham. https://doi-org.printkr.hs-niederrhein.de:2443/10.1007/978-3-031-45282-6_6, accessed on 10.04.2024.

Security issue	Description	Source
Leaking data	ChatGPT lacks encryption, strict access control, and access logs. This is similar to the use of git repositories, where sensitive files can often end up despite the lack of sufficient security controls. More than 4% of employees have put sensitive corporate data into ChatGPT, raising security fears	Powell (2023)
Phishing and social engineering attacks	ChatGPT can be used to create convincing personas to steal information and create phishing emails. Without proper security education and training, ChatGPT users could inadvertently put sensitive information at risk	Shah (2023)
Malware	ChatGPT can be used to create malware	Leong (2023) Vijayan (2023)
Biased behavior	ChatGPT can sometimes respond to harmful instructions or exhibit biased behavior	Sangfor (2023)
Trickery	ChatGPT is remarkably easy to trick	Harrison (2023)
Manipulating user trust and deception	The model's ability to generate realistic and contextually relevant responses can be exploited to deceive users. Attackers may use ChatGPT to craft messages that appear legitimate, providing false information or misleading instructions. Such deceptive tactics can be employed for various malicious purposes	Shalabaieva (2023)
Lack of access control	ChatGPT lacks strict access control	Jackson and McDaniel (2023)
Data leaks from ChatGPT	ChatGPT often gives an example of completely insecure code	Jackson and McDaniel (2023) Umawing (2023), James (2023)
Fake Website	ChatGPT can generate text that closely resembles authentic web content, making it easier for attackers to create deceptive websites. These fake websites can be used to collect sensitive information from unsuspecting users, such as login credentials or financial data. Additionally, ChatGPT's ability to mimic human-like conversations opens up possibilities for impersonation, where attackers can manipulate users by posing as trusted individuals or entities	Kargl (2023)
Deepfakes attack	The Misuse of ChatGPT's language generation capabilities to fabricate believable but false text, such as realistic-seeming messages, articles, or posts. These deepfakes can be used to deceive, spread disinformation, or impersonate individuals in digital conversations. Just like deepfake videos, where AI is used to create hyper-realistic but fake video content	Vedova and Atleson (2023)

Figure 10: “Non-exhaustive list of ChatGPT security and privacy issues.”¹⁹⁴

Security issue	Description	Source
Model poison attack	A ChatGPT model poisoning attack refers to a situation where the training data used to educate the AI system is deliberately manipulated with malicious intent. This could involve injecting biased, misleading, or harmful information into the training data. The resulting model would then generate outputs that mirror this skewed input, leading to harmful, biased, or false outputs. This kind of attack can undermine the reliability of the AI and can have serious consequences, especially if the system is used in decision-making processes	Palmer (2023)
Model API attack	A ChatGPT API attack refers to a situation where a malicious actor tries to exploit the application programming interface (API) that provides access to the ChatGPT model. This could be an attempt to gain unauthorized access, manipulate the model's outputs, or overload the system to cause a denial of service. Such attacks can lead to misuse of the system, leakage of sensitive data, or disruption of the service, and hence pose significant security concerns	Casillo and Powell (2023)
Prompt injection attack	Prompt injection against ChatGPT refers to a potential attack where a malicious actor deliberately crafts specific prompts designed to make the model generate harmful, biased, or misleading responses. Such attacks exploit the AI's dependency on input prompts to guide its responses. If successful, the attacker could manipulate the system to produce content that aids misinformation campaigns, fosters bias, or otherwise causes harm	Zhang (2023)
Hallucination attacks	ChatGPT hallucinates by its very nature and can generate links that can then be used for typosquatting. The same thing with package managers etc.	Lanyado (2023)

Figure 10 (continued)¹⁹⁵

6.3 Ethical Guidelines for Chat GPT Usage in Healthcare

The use of AI in the healthcare sector gives rise to a range of ethical and legal concerns. Historically, healthcare experts were solely responsible for making healthcare choices. The utilization of intelligent technologies, which independently make judgments or aid in diverse activities, prompts inquiries over accountability, openness, consent, and secrecy.¹⁹⁶

The first obstacle to tackle is undeniably transparency, given that the majority of AI algorithms are inherently complex and inscrutable.

¹⁹⁴ Huang, K., Duan, Y. (2023). ChatGPT in Nutrition Science. In: Huang, K., Wang, Y., Zhu, F., Chen, X., Xing, C. (eds) Beyond AI. Future of Business and Finance. Springer, Cham. https://doi-org.printkr.hs-niederrhein.de:2443/10.1007/978-3-031-45282-6_6, accessed on 10.04.2024.

¹⁹⁵ Huang, K., Duan, Y. (2023). ChatGPT in Nutrition Science. In: Huang, K., Wang, Y., Zhu, F., Chen, X., Xing, C. (eds) Beyond AI. Future of Business and Finance. Springer, Cham. https://doi-org.printkr.hs-niederrhein.de:2443/10.1007/978-3-031-45282-6_6, accessed on 10.04.2024.

¹⁹⁶ Cf. Romagnoli, A., Ferrara, F., Langella, R. et al. Healthcare Systems and Artificial Intelligence: Focus on Challenges and the International Regulatory Framework. Pharm Res (2024). <https://doi-org.printkr.hs-niederrhein.de:2443/10.1007/s11095-024-03685-3>, accessed on 10.04.2024.

The second issue pertains to the moral duty, namely the need to acknowledge accountability for one's conduct, which becomes challenging to delineate in the context of AI use. Both the creator and the operator who chooses to use artificial intelligence might be ascribed with a moral duty. The World Health Organization (WHO) has established rules to manage and govern the use of AI in healthcare as outlined in Figure 11.

These principles aim to minimize dangers and optimize the potential of AI in healthcare.¹⁹⁷

Protect human autonomy	• People should govern health care systems and medical decisions. Privacy and confidentiality should be protected, and patients should give informed consent using appropriate legal frameworks for data protection
Promote Human Welfare and Security, as well as the Public Interest	• Regulatory standards related to safety and effectiveness must be met by AI technology designers.
Importanza della Trasparenza e Comprensibilità	• It is necessary to publish a sufficient amount of data before the implementation of AI technology.
Promote Responsibility	• Operators must ensure that it is used under the proper conditions and by trained personnel.
Ensuring Inclusion and Equity	• It needs to be designed to promote its use and access regardless of age, gender, sex, income, race, sexual orientation
Sustainability	• Algorithms should be designed to have the least environmental impact and use the least amount of feasible energy

Figure 11: “Fundamental ethical principles for the correct management of AI in healthcare.”¹⁹⁸

The primary obstacles to proper use of AI in healthcare with crucial factors that stem from existing machine learning (ML) methods, which significantly depend on training data. Even with unsupervised learning, which is supposedly autonomous from direct human monitoring, a relationship to supervision procedures might nevertheless arise. These issues create important concerns about

¹⁹⁷ Cf. Romagnoli, A., Ferrara, F., Langella, R. et al. Healthcare Systems and Artificial Intelligence: Focus on Challenges and the International Regulatory Framework. Pharm Res (2024). <https://doi-org.printkr.hs-niederrhein.de/2443/10.1007/s11095-024-03685-3>, accessed on 10.04.2024.

¹⁹⁸ Ebd.

the accuracy, protection, and confidentiality of data, which emphasizes the need for thorough investigation.¹⁹⁹

When it comes to public health research practice, big data presents three main concerns: the possibility of accidentally revealing personal information, for example through online tools, the potential complexity of determining if personal information can be deduced from the data, and the difficulty of upholding ethical research standards in the presences of new technologies that can change established privacy norms such as GPS, drones, and social media. The importance of data quality becomes evident in guaranteeing the accuracy of machine learning models, as the dependability of outcomes is directly linked to the quality of data used throughout the training phase.²⁰⁰

Another area of concern is to the possible danger associated with statistically biased or inaccurate models. It's necessary to do audits on the 'black box,' meaning to comprehend the internal mechanisms of models. Moreover, it's necessary to enhance system interoperability, with the goal of guaranteeing seamless cooperation and compatibility between them. Implementing this method is essential for reducing risks and enhancing transparency in the use of ML/DL models in healthcare settings.²⁰¹

6.4 Addressing Bias and Fairness Issues in Chat GPT Algorithms

The use of GPT technology in the healthcare sector, like any novel technology, presents significant ethical concerns and challenges that need thorough deliberation. Several prominent ethical concerns and challenges arise while using GPT in the health care sector:

- 1) Data privacy: GPT models require access to substantial quantities of textual data in order to achieve good training. This gives rise to apprehensions over the protection of patient confidentiality and the possibility of unintentional disclosure of sensitive data. It is crucial to establish adequate safeguards to safeguard patient data and to guarantee that patients are well informed about the use of their data. Essentially, the

¹⁹⁹ Ebd.

²⁰⁰ Cf. Romagnoli, A., Ferrara, F., Langella, R. et al. Healthcare Systems and Artificial Intelligence: Focus on Challenges and the International Regulatory Framework. *Pharm Res* (2024). <https://doi-org.printkr.hs-niederrhein.de/2443/10.1007/s11095-024-03685-3>, accessed on 10.04.2024.

²⁰¹ Ebd.

security level of the “firewall” protecting patient data should be equivalent to the degree of protection used in online banking.²⁰²

- 2) Bias: GPT models have the ability to mirror the biases that exist in the data they are trained on. This may lead to biased consequences for patients depending on variables such as race or gender. Thoroughly evaluating the choice and organization of data used for training GPT models is crucial, as is doing bias assessments on the resultant models.²⁰³
- 3) Accountability: With the increasing sophistication of GPT models, they are capable of producing content that is difficult to differentiate from human-written material. This gives rise to problems about the responsibility for judgments taken on the basis of text created by GPT. Establishing explicit criteria for the use of GPT-generated text in decision-making is essential, as is ensuring a transparent chain of author accountability for any choices made within this framework.²⁰⁴
- 4) Informed consent: Patients must be informed that their data will be used for training GPT models and hence must have a comprehensive understanding of the possible consequences of its usage in such a setting. Prior to using patient data to train GPT models, it's important to get informed permission about the use of GPT technique.²⁰⁵
- 5) Opacity: GPT models may exhibit opacity, indicating that comprehending the process behind a certain choice or the creation of a particular text segment might be challenging. It's necessary to provide transparency in GPT models and ensure that the decision-making process can be explained to patients and healthcare professionals. Although GPT technology has the potential to enhance patient outcomes and efficiency in healthcare, it's vital to carefully assess the ethical aspects of its usage and take required measures to prevent any dangers and concerns.²⁰⁶

It is essential to establish rules for the use of Chat GPT in scientific research, taking into consideration the concerns of responsibility, ethical conduct, openness,

²⁰² Cf. Kumaresan, A., Uden, L., Ashraf, S. (2023). Potential Role of ChatGPT in Healthcare in the Prevention and Management of Non-communicable Diseases. In: Uden, L., Ting, I.H. (eds) Knowledge Management in Organisations. KMO 2023. Communications in Computer and Information Science, vol 1825. Springer, Cham. https://doi-org.printkr.hs-niederrhein.de:2443/10.1007/978-3-031-34045-1_34, accessed on 10.04.2024.

²⁰³ Cf. Kumaresan, A., Uden, L., Ashraf, S. (2023). Potential Role of ChatGPT in Healthcare in the Prevention and Management of Non-communicable Diseases. In: Uden, L., Ting, I.H. (eds) Knowledge Management in Organisations. KMO 2023. Communications in Computer and Information Science, vol 1825. Springer, Cham. https://doi-org.printkr.hs-niederrhein.de:2443/10.1007/978-3-031-34045-1_34, accessed on 10.04.2024.

²⁰⁴ Ebd.

²⁰⁵ Ebd.

²⁰⁶ Ebd.

and truthfulness. Utilizing Chat GPT in the field of healthcare necessitates a meticulous and conscientious approach, taking into account the possible hazards and apprehensions it presents.²⁰⁷

Establishing a legal framework for the use of Chat GPT is vital to provide adequate protection in research and validation processes. Mira Murati, the Chief Scientific Officer of Chat GPT supports the implementation of regulations to address and reduce the potential for harmful and malicious use of the technology.²⁰⁸

Currently, there are no enforceable U.S. regulatory limitations on generative AI in the USA. However, the Biden Administration's Blueprint for an AI Bill of Rights outlines principles for responsible deployment of AI, which includes advocating for rights against 'algorithmic discrimination'. The EU seeks to establish regulations governing the use of AI in both commercial and governmental contexts, including generative AI, sometimes referred to AI ACT. Glorin Sebastian argues that authorities should enforce data protection regulations to safeguard users' privacy by controlling the gathering and use of data produced by the AI system. Sebastian proposes using intellectual property laws to prevent chatbot systems from using copyrighted content without permission.²⁰⁹

7. Implementing Chat GPT in the Healthcare Sector

Healthcare systems are now facing the task of addressing the rapid increase in healthcare expenses globally, which has exceeded the growth rate of Gross Domestic Product (GDP) in individual states. The limited financial resources, together with growing elderly population and a rise in chronic illness, may pose a threat to their long-term viability.²¹⁰

The significance of AI in healthcare and its capacity to tackle healthcare delivery challenges has been a topic of discourse within the scientific community for several years. The use of AI in healthcare is expected to have significant benefits, as these technologies have shown promise in various medical fields, such as mental health, radiology, oncology, and ophthalmology. Additionally, AI can help

²⁰⁷ Ebd.

²⁰⁸ Cf. Kumaresan, A., Uden, L., Ashraf, S. (2023). Potential Role of ChatGPT in Healthcare in the Prevention and Management of Non-communicable Diseases. In: Uden, L., Ting, I.H. (eds) Knowledge Management in Organisations. KMO 2023. Communications in Computer and Information Science, vol 1825. Springer, Cham. https://doi-org.printkr.hs-niederrhein.de:2443/10.1007/978-3-031-34045-1_34, accessed on 10.04.2024.

²⁰⁹ Ebd.

²¹⁰ Cf. Romagnoli, A., Ferrara, F., Langella, R. et al. Healthcare Systems and Artificial Intelligence: Focus on Challenges and the International Regulatory Framework. Pharm Res (2024). <https://doi-org.printkr.hs-niederrhein.de:2443/10.1007/s11095-024-03685-3>, accessed on 10.04.2024.

in reducing waitlists, improving treatment adherence, and personalizing therapy. Key stakeholders are advocating for a tangible incorporation of AI into clinical practice, beyond the theoretical or experimental limitations previously described.²¹¹

Although the Food and Drug Administration (FDA) has authorized a growing number of clinical algorithms based on artificial intelligence (AI), their actual application in real-world settings remains limited. The difficulties faced in the process of incorporating AI into clinical practice extend well beyond the first stages of creation and evaluation.²¹²

Therefore, it is essential for the scientific community to possess a comprehensive of AI-based medical technologies and their regulatory framework, in order to evaluate their inclusion in the arsenal of tools accessible to healthcare providers.²¹³

Chat GPT enables healthcare practitioners to give symptom assessments and virtual patient assistance, while also addressing typical medical queries.

Integrating Chat GPT into healthcare would allow a chatbot to provide an initial diagnosis by analyzing a patient's symptoms. This might assist the user in comprehending their situation and proposing potential further actions.²¹⁴

7.1 Integrating Chat GPT into Existing Healthcare Systems



Figure 12: “Opportunities and limitations of medical LLM-based systems.”²¹⁵

Figure 12 provides an in-depth analysis of the most favorable prospects and potential drawbacks associated with the use of LLMS in the healthcare sector.

²¹¹ Ebd.

²¹² Cf. Romagnoli, A., Ferrara, F., Langella, R. et al. Healthcare Systems and Artificial Intelligence: Focus on Challenges and the International Regulatory Framework. Pharm Res (2024). <https://doi-org.printkr.hs-niederrhein.de:2443/10.1007/s11095-024-03685-3>, accessed on 10.04.2024.

²¹³ Ebd.

²¹⁴ Cf. Waghmare, C. (2023). Introduction to ChatGPT. In: Unleashing The Power of ChatGPT. Apress, Berkeley, CA. https://doi-org.printkr.hs-niederrhein.de:2443/10.1007/979-8-8688-0032-0_1, accessed on 13.04.2024.

²¹⁵ García-Méndez, S., de Arriba-Pérez, F. Large Language Models and Healthcare Alliance: Potential and Challenges of Two Representative Use Cases. Ann Biomed Eng(2024). <https://doi-org.printkr.hs-niederrhein.de:2443/10.1007/s10439-024-03454-8>, accessed on 13.04.2024.

LLMS are able to effectively analyze large quantities of data due to their exceptional filtering capabilities. Therefore, they show significant promise, enhancing the efforts of doctors, particularly when clinical calculations are projected to grow using historical data.²¹⁶

Medical report creation benefits from their natural language generating skills in a similar manner. It's important to consider using question-answering activities with these systems to encourage active participation and ensure that interventions are followed and effective.²¹⁷

The latter is also associated with personalized healthcare, an urgent need in the medical domain that becomes achievable with LLMS by analyzing users' everyday routines, lifestyle, and stress levels, along with other pertinent elements. Specifically, the hazards associated with using LLMS in the healthcare sector include cognitive decline, post-partum depression, chronic diseases and several other medical applications.²¹⁸

One should take into account the lack of transparency in the models, meaning the absence of clear reasoning behind how they work. This lack of openness might have a detrimental influence on the end-users, namely practitioners and patients. To enhance model interpretability, one may integrate physical principles into the system and use rLHF to guarantee fairness.²¹⁹

Addressing low accuracy issues may be achieved by including the probabilistic certainty of their answer. LLMS has many advantages for detecting and monitoring cognitive decline and postpartum depression, including anonymity and cost reduction, which are particularly notable when compared to conventional methods. Prior to incorporating them into healthcare systems, it is essential to address problems notwithstanding their promising performance.²²⁰

Initially, the protection of data privacy and the accuracy of training data including sensitive medical information pose significant difficulties. Additionally, the continuous updating of medical literature adds to the complexity. The absence of interpretability in intelligent medical systems is a very concerning issue, as discussed before. During the age of LLMS, it is of the utmost significance to prioritize establishing confidence in AI in order to address the black box dilemma.

²¹⁶ Ebd.

²¹⁷ Cf. García-Méndez, S., de Arriba-Pérez, F. Large Language Models and Healthcare Alliance: Potential and Challenges of Two Representative Use Cases. *Ann Biomed Eng* (2024), <https://doi-org.printkr.hs-niederrhein.de/2443/10.1007/s10439-024-03454-8>, accessed on 13.04.2024.

²¹⁸ Ebd.

²¹⁹ Ebd.

²²⁰ Ebd.

LLM-based medical solutions currently have limits in terms of fairness, as they may demonstrate biased performance depending on demographic variables like as race, gender and socioeconomic level. These limitations arise from the use of domain or local data, which may affect their generalizability and real-world performance.²²¹

Ultimately, the General Data Protection Regulation should be implemented using innovative privacy methods for data access and storage that guarantee data security while still allowing for sophisticated data analysis.²²²

7.2 Training Healthcare Professionals to Collaborate with Chat GPT

The future of Chat GPT in healthcare has significant potential via collaboration, and digital interaction with healthcare professionals. Engaging healthcare professionals in the development and use of Chat GPT ensures that the AI system is tailored to their requirements and enhances the quality of patient care. Major elements of this partnership comprise²²³:

- Feedback and improvement: Input from healthcare experts helps in refining and enhancing Chat GPT. Their observations on performance, usability, and clinical significance allow incremental, improvements, customizing the technology to fit healthcare processes.²²⁴
- Domain-specific customization: Healthcare experts may contribute to the customization of Chat GPT specifically for the healthcare industry. This partnership assists in addressing precise healthcare circumstances and enables more precise interactions.²²⁵
- Healthcare experts' participation in the validation and assessment of Chat GPT is essential. The effectiveness, and safety of the product are evaluated by rigorous testing, in which the users actively participate, in real-world healthcare settings.²²⁶
- The co-design to interfaces involves collaborative design efforts to produce user-friendly interfaces that are tailored to the current workflows

²²¹ Ebd.

²²² Cf. García-Méndez, S., de Arriba-Pérez, F. Large Language Models and Healthcare Alliance: Potential and Challenges of Two Representative Use Cases. *Ann Biomed Eng* (2024), <https://doi-org.printkr.hs-niederrhein.de/2443/10.1007/s10439-024-03454-8>, accessed on 13.04.2024.

²²³ Cf. Benet, David. "ChatGPT/AI in Healthcare Management." *Journal of Clinical Medical Research* (2023): n. pag. <http://dx.doi.org/10.46889/JCMR.2023.4301>, accessed on 15.04.2024.

²²⁴ Ebd.

²²⁵ Ebd.

²²⁶ Ebd.

of healthcare practitioners. This guarantees the smooth incorporation of Chat GPT into clinical procedures and effective communication.²²⁷

- Healthcare professionals have the ability to contribute to the ethical issues and recommendations throughout the development of Chat GPT. Their observations contribute to the maintenance of patient confidentiality, the obtaining of informed agreement, and the exercise of ethical judgment.²²⁸
- Education and training: Collaborative educational and training initiatives provide healthcare professionals with the necessary knowledge and skills to properly employ Chat GPT and make well-informed choices base on its generated results.²²⁹
- Real-time feedback and continuous improvement: mechanisms for real-time feedback facilitate continued partnerships between healthcare experts and developers. Frequent contact enables them to provide comments, report concerns, and propose improvements for future revisions of Chat GPT.²³⁰

Through the active participation of healthcare professionals in the cooperation and co-creation process, they can use their experience to influence the development and enhancement of Chat GPT. This strategy guarantees that it is in line with the requirements of healthcare professionals, hence improving patient cate, clinical decision-making, and the entire healthcare experience.²³¹

7.3 Ensuring Continuous Improvement and Updating of Chat GPT Algorithms

The combination of findings from the extensive GBD research and the sophisticated AI capabilities of open AI Chat GPT-4 has the potential to revolutionize customized healthcare planning. By combining the data-driven insights from the GBD research with the advanced conversational skills of Chat GPT-4, healthcare practitioners may create personalized healthcare strategies that align with patients' lives and preferences. Researchers suggest that this

²²⁷ Ebd.

²²⁸ Cf. Benet, David. "ChatGPT/AI in Healthcare Management." *Journal of Clinical Medical Research* (2023): n. pag. <http://dx.doi.org/10.46889/JCMR.2023.4301>. accessed on 15.04.2024.

²²⁹ Ebd.

²³⁰ Ebd.

²³¹ Ebd.

groundbreaking collaboration has the potential to develop a unique AI-supported tool for assessing and planning personalized disease burden.²³²

In order to effectively apply this non-traditional technology, it is essential to guarantee consistent and precise updates, expert oversight, and address any possible biases and restrictions. Healthcare professionals and stakeholders should adopt a well-rounded and flexible approach, giving importance to partnerships across different disciplines, ensuring accurate data, promoting openness, adhering to ethical standards, and continuously updating their skills via training.²³³

To increase individualized healthcare planning, they may invest in the distinctive capabilities of both Chat GPT-4, particularly its recently announced features like live internet surfing or plugins, and the GBD research. This novel method has the capacity to enhance patient results and optimize the use of resources, while also setting the stage for the global adoption of precision medicine, thereby transforming the current healthcare environment.²³⁴

However, in order to fully use these advantages on a worldwide scale and for individuals, further research and development are necessary. By doing this they will be able to fully use the potential of this collaboration, which will result in societies moving closer to a future where individualized treatment is the standard rather than the rare occurrence.²³⁵

8. Future Directions and Possibilities for Chat GPT in Chronic Disease Management

Chatbots have gained popularity in the management of chronic illnesses due to its ability to provide personalized assistance and interactions, while also being convenient and accessible for patients to manage their conditions.²³⁶

This subchapter examines the capabilities of Chat GPT, an advanced AI system, and analyze its potential in assisting persons with chronic diseases via the provision of individualized guidance, symptom monitoring, and lifestyle suggestions.

²³² Cf. Tamsah, Mohamad-Hani et al. "ChatGPT-4 and the Global Burden of Disease Study: Advancing Personalized Healthcare Through Artificial Intelligence in Clinical and Translational Medicine." *Cureus* 15 (2023): n. pag. <https://api.semanticscholar.org/CorpusID:258863204>, accessed on 15.04.2024.

²³³ Cf. Tamsah, Mohamad-Hani et al. "ChatGPT-4 and the Global Burden of Disease Study: Advancing Personalized Healthcare Through Artificial Intelligence in Clinical and Translational Medicine." *Cureus* 15 (2023): n. pag. <https://api.semanticscholar.org/CorpusID:258863204>, accessed on 15.04.2024.

²³⁴ Ebd.

²³⁵ Ebd.

²³⁶ Cf. Haque, Ahshanul; Chowdhury, Md Naseef-Ur-Rahman; Soliman, Hamdy (2023). Transforming Chronic Disease Management with Chatbots: Key Use Cases for Personalized and Cost-effective Care. TechRxiv. Preprint. <https://doi.org/10.36227/techrxiv.22579735.v2>, accessed on 15.04.2024.

8.1 Advancements in Natural Language Processing and AI for Healthcare

A) Personalized education: Chatbots may provide personalized instruction to patients with chronic illness, helping them get better understanding of their condition and effectively manage its symptoms. Chatbots may use AI and NLP to evaluate a patients knowledge level and customize their education based on their specific needs and preferences. Chatbots may also provide educational resources in various formats, like as recordings, articles, and interactive assessments, to engage with patients and enhance their prospects for personal development.²³⁷

Individualized education may strengthen patients' health literacy and self-efficacy, enabling them to actively participate in their own treatment.

Individualized education may be advantageous for patients with chronic illness in enhancing medication adherence, self-care routines, and clinical results.²³⁸

Chatbots may also provide continuous training and support to patients, helping them maintain their knowledge and skills over an extended period of time.²³⁹

However, there are certain challenges associated with personalized education via chatbots. For example, chatbots will not be able to address all of the patient's interests and queries, and they are unlikely to provide personalized feedback on their progress. Additionally, patients may possess diverse learning methods and preferences, which may not be fully addressed by chatbots. It is vital to design chatbots that are adaptive and versatile, capable of providing a wide range of educational materials and resources to fulfill the diverse needs of patients.²⁴⁰

Chatbots that provide individualized instruction have the potential to increase the effectiveness of chronic illness management and better patient outcomes. Chatbots may enhance patients' understanding of their illness, facilitate informed decision-making about their treatment, and contribute

²³⁷ Cf. Haque, Ahshanul; Chowdhury, Md Naseef-Ur-Rahman; Soliman, Hamdy (2023). Transforming Chronic Disease Management with Chatbots: Key Use Cases for Personalized and Cost-effective Care. TechRxiv. Preprint. <https://doi.org/10.36227/techrxiv.22579735.v2>, accessed on 15.04.2024.

²³⁸ Ebd.

²³⁹ Ebd.

²⁴⁰ Ebd.

to improved health outcomes by offering personalized education and continuous support.²⁴¹

- B) Symptom Monitoring: Chatbots may aid patients with chronic diseases in monitoring their symptoms and tracking their progress over a period of time. Patients have the ability to use Chatbots to promptly communicate their problems, get personalized advice, and make adjustments to their treatment regimens accordingly. Patients may get notifications and prompts from chatbots to ensure they take their prescribed prescriptions, monitor their blood glucose levels, and attend scheduled appointments. Researchers have seen similar applications in the field of mobile health.²⁴² Chatbot-based symptom monitoring may enhance patient outcomes by encouraging self-management, mitigating complications, and promptly identifying and addressing symptoms. Furthermore, the process of monitoring side effects may include the exchange of information between patients and healthcare professionals, as well as efforts to enhance patient satisfaction and engagement.²⁴³

However, there are some challenges associated with monitoring adverse effects using chatbots. Erroneous evaluations and recommendations may occur when patients provide inconsistent or erroneous reports of their symptoms. Furthermore, chatbots may lack the ability to identify intricate symptoms or tiny fluctuations in a patient's state, thereby requiring more advanced monitoring and evaluation techniques.²⁴⁴

Chatbot-based symptom monitoring has the potential to improve the overall quality of treatment and results for patients with chronic conditions. Chatbots may aid patients in efficiently controlling their symptoms, averting problems, and attaining improved health results by offering customized feedback and assistance.²⁴⁵

- C) Pharmaceutical Management: Chatbots may assist individuals with chronic diseases in managing their drug regimen. Chatbots may provide timely medication reminders to patients, check their medicine consumption, and inform them about possible drug interactions or adverse

²⁴¹ Ebd.

²⁴² Cf. Haque, Ahshanul; Chowdhury, Md Naseef-Ur-Rahman; Soliman, Hamdy (2023). Transforming Chronic Disease Management with Chatbots: Key Use Cases for Personalized and Cost-effective Care. TechRxiv. Preprint. <https://doi.org/10.36227/techrxiv.22579735.v2>, accessed on 15.04.2024.

²⁴³ Ebd.

²⁴⁴ Ebd.

²⁴⁵ Ebd.

effects. Chatbots may aid patients in the process of renewing or obtaining refills for their medicines and provide instructional resources on the dose, administration, and use of their drugs.²⁴⁶

Effective medication management is important to achieving optimum health outcomes in individuals with chronic illnesses. Poor drug adherence may exacerbate a condition, require hospitalization, and finish quality of life. Chatbots may enhance medication adherence and patient safety by monitoring drug use, offering individualized reminders and education, and notifying patients and healthcare professionals about possible concerns.²⁴⁷

There are certain challenges associated with providing medical assistance to CEOs via Chatbots. Chatbots may lack the capacity to adequately cater to the diverse medication regimes, doses, and preferences of patients. It is conceivable that chatbots may fail to identify instance of non-adherences caused by forgetfulness, deliberate non-compliance, or other factors.

Pharmaceutical Management chatbots may improve patient safety, medication adherence, and the efficacy of managing chronic illnesses. Chatbots may aid patients in efficiently managing their prescriptions, preventing problems, and attaining improved health outcomes via the provision of individualized information and reminders.²⁴⁸

- D) Lifestyle Coaching: Chatbots can offer provide lifestyle coaching to patients with chronic illnesses, assisting them in adopting good lifestyle choices and engaging in certain behaviors. Chatbots utilize AI and NLP to assess a patient's lifestyle patterns and offer tailored guidance and feedback to improve their dietary choices, physical activity, sleep, stress management, and other health-related behaviors. Chatbots have the capability to monitor the development of patients over a period of time and provide motivation, support, and accountability.²⁴⁹

Chatbot-based lifestyle coaching may empower patients to adopt enduring lifestyle changes, mitigate their risk of problems, and enhance their overall well-being. Research has shown that using chatbot-based lifestyle coaching may enhance several aspects of health, including as food,

²⁴⁶ Ebd.

²⁴⁷ Cf. Haque, Ahshanul; Chowdhury, Md Naseef-Ur-Rahman; Soliman, Hamdy (2023). Transforming Chronic Disease Management with Chatbots: Key Use Cases for Personalized and Cost-effective Care. TechRxiv. Preprint. <https://doi.org/10.36227/techrxiv.22579735.v2>, accessed on 15.04.2024.

²⁴⁸ Ebd.

²⁴⁹ Ebd.

physical activity, weight, blood pressure, and other health indicators, in patients with chronic illness. Lifestyle coaching may also enhance patient engagement and contentment with healthcare.²⁵⁰

There are certain challenges associated with lifestyle coaching via chatbots. Individualized coaching tactics may be necessary due to variations in patient preferences, motivations, and challenges to behavior change. There is a possibility that chatbots may not have capability to address all the issuers of the patient or provide personalized feedback on their progress. It is important to develop chatbots that are flexible, quick to respond, and able to provide a range of coaching techniques and resources to meet the diverse need of patients.²⁵¹

Using chatbots for lifestyle training has the potential to enhance the comprehension of outcomes and enhance the effectiveness of managing chronic illnesses. Chatbots may aid patients in adopting good living habits, mitigating complications, and attaining improved health outcomes via the provision of personalized coaching and support.²⁵²

8.2 Potential Impact of Chat GPT on Chronic Disease Management Outcomes

This subchapter explores the potential impact of Chat GPT on chronic disease management and how it can support individuals with chronic conditions by giving tailored guidance, tracking symptoms, and offering lifestyle suggestions.

Patient Engagement refers to the active involvement of patients in their own healthcare. It involves patients taking an active role in making decisions about their treatment, managing their health conditions, and participating in healthcare activities.

Chatbots have the enhance patient engagement in controlling chronic illnesses, which is a very advantageous feature. Dedication is vital for the advancement of chronic illness management since it may be necessary to actively participate in their well-being in order to achieve optimal outcomes. Chatbots have several methods to enhance engagement.²⁵³

²⁵⁰ Ebd.

²⁵¹ Cf. Haque, Ahshanul; Chowdhury, Md Naseef-Ur-Rahman; Soliman, Hamdy (2023). Transforming Chronic Disease Management with Chatbots: Key Use Cases for Personalized and Cost-effective Care. TechRxiv. Preprint. <https://doi.org/10.36227/techrxiv.22579735.v2>, accessed on 15.04.2024.

²⁵² Ebd.

²⁵³ Ebd.

Initially, individuals may get tailored guidance, instruction, and assistance from chatbots, which can augment their self-confidence and their understanding of their medical condition.²⁵⁴

Chatbots may engage patients by communicating fitted facts and providing instructions, enabling them to take responsibility for their health, set achievable goals, and track their progress toward achieving them.²⁵⁵

They may aid patients in communicating more effectively with their healthcare professionals. And provide them with real-time access to healthcare information and services, enabling them to keep informed about their treatment, ask questions, and report any concerns or symptoms. Consequently, healthcare personnel may enhance their decision-making process in relation to patient care.²⁵⁶

Chatbots have the capability to enhance patients' involvement and compliance with treatment regimens by offering continuous assistance and incentive. They possess the capacity to aid patients in maintaining adherence to their treatment regimen and surmounting any obstacles or impediments by dispatching reminders, monitoring advancement, and providing motivation.²⁵⁷

Using chatbots to enhance patient engagement has some challenges. Chatbots' deficiency in empathy and emotional intelligence may lead to decrease in patients' trust and involvement. In addition, it is important to note that chatbots may not be beneficial for all patients, especially those with severe or complicated health issues that need considerable human interaction.²⁵⁸

It is important to develop chatbots that are flexible, reactive, and able to provide a range of interaction methods and resources to meet the diverse needs of patients. By using meticulous design and execution, chatbots has the capability to amplify patient involvement and optimize the effectiveness of chronic illness management.²⁵⁹

Cost-effective care utilizing chatbots in the treatment of chronic illnesses offers a significant benefit in term of cost-effectiveness. The exorbitant expenses linked to hospitalization, emergency room visits, and medicines for chronic illnesses exert a substantial burden on the healthcare system. Chatbots have the potential to reduce

²⁵⁴ Ebd.

²⁵⁵ Ebd.

²⁵⁶ Cf. Haque, Ahshanul; Chowdhury, Md Naseef-Ur-Rahman; Soliman, Hamdy (2023). Transforming Chronic Disease Management with Chatbots: Key Use Cases for Personalized and Cost-effective Care. TechRxiv. Preprint. <https://doi.org/10.36227/techrxiv.22579735.v2>, accessed on 15.04.2024.

²⁵⁷ Ebd.

²⁵⁸ Ebd.

²⁵⁹ Ebd.

healthcare expenses by providing patients with access to care and resources without the need for in-person visits.²⁶⁰

Chatbots may provide cost-efficient healthcare via several means. Firstly, chatbots may assist patients in remotely managing their health, hence minimizing the need for in-person visits to healthcare institutions. This may lead to cost savings for both patients and healthcare professionals, since it reduces the need for travel, taking time off work, and other related expenses.²⁶¹

Furthermore, chatbots have the capability to avert hospitalizations and trips to the emergency room by offering sufferers immediate monitoring of their symptoms, adherence to medicine, and vital signs. This may assist healthcare providers in identifying possible difficulties at the early stage and intervening before they worsen, hence decreasing the need for expensive hospitalizations and emergency room visits.²⁶²

Chatbots have the capability to provide sufferers continuous education and assistance, therefore aiding in the prevention of illness advancement and reducing the need for costlier remedies in the future. They may assist patients in efficiently managing their chronic diseases by encouraging good lifestyle practices and ensuring medication adherence. This can ultimately lead to a decrease in the need for more expensive treatments over time.²⁶³

Nevertheless, there exist some obstacles linked to use of chatbots for delivering cost-efficient healthcare. Healthcare providers may face obstacles in using chatbots due to the need for early investment in technology and infrastructure. Moreover, chatbots may not be appropriate for every patient, especially those with intricate or severe health concerns that need more extensive and individualized treatment.²⁶⁴

It becomes essential to thoroughly assess the potential financial benefits linked to chatbots in the treatment of chronic diseases and to evaluate their efficacy and cost-effectiveness in real-world medical settings. Through proper design and execution, chatbots have the capacity to provide cost-efficient treatment to those with chronic illnesses and alleviate the financial strain on the Healthcare system.²⁶⁵

²⁶⁰ Ebd.

²⁶¹ Ebd.

²⁶² Cf. Haque, Ahshanul; Chowdhury, Md Naseef-Ur-Rahman; Soliman, Hamdy (2023). Transforming Chronic Disease Management with Chatbots: Key Use Cases for Personalized and Cost-effective Care. TechRxiv. Preprint. <https://doi.org/10.36227/techrxiv.22579735.v2>, accessed on 15.04.2024.

²⁶³ Ebd.

²⁶⁴ Ebd.

²⁶⁵ Ebd.

8.3 Exploring Other Applications of Chat GPT in the Healthcare Sector

The promise of Chat GPT on the healthcare industry goes beyond its core emphasis on improving chronic illness management. This Bachelor Thesis largely examines the ways in which Chat GPT might assist persons with chronic diseases by giving tailored advice, monitoring symptoms, and offering lifestyle suggestions. However, there are also more potential uses that are worth considering. This subchapter explores many different applications, providing insight into the numerous ways in which Chat GPT might transform Healthcare. An area with potential for use of Chat GPT in the provision of mental health assistance. Chronic mental health disorders, such as depression and anxiety often need continuous surveillance and tailored therapies. Chat GPT may serve as a virtual therapist, providing a secure environment for people to express their issues and provide coping skills based on scientific facts. Furthermore, Chat GPT has the capability to observe fluctuations in mood and behavioral trends over a period of time, notifying healthcare personnel of any emergencies and facilitating prompt actions.²⁶⁶

Another potential use for this technology is in the field of preventative healthcare. Chat GPT may provide tailored suggestions for illness prevention by examining a patient's medical records, lifestyle decisions, and genetic inclinations.²⁶⁷ Chat GPT has the ability to provide specific recommendations for test, offer dietary modifications, and encourage exercise routines.²⁶⁸ This empowers users to take proactive measures in order to preserve their health and avoid the occurrence of chronic illness.²⁶⁹

Moreover, Chat GPT may play an important role in enhancing adherence to medication. A significant number of people have difficulties in adhering to their prescribed drug regimens, resulting in negative health consequences. Chat GPT may function as a medicine reminder, delivering prompt notifications and addressing inquiries on dose, side effects, and possible interactions. Chat GPT

²⁶⁶ Cf. Alanezi, Fahad. "Assessing the Effectiveness of ChatGPT in Delivering Mental Health Support: A Qualitative Study." *Journal of Multidisciplinary Healthcare* 17 (2024): 461 – 471, <https://api.semanticscholar.org/CorpusID:267373431>, accessed on 16.04.2024.

²⁶⁷ Cf. Zhou, Zeyu. "Evaluation of ChatGPT's Capabilities in Medical Report Generation." *Cureus* 15 (2023): n. pag., <https://api.semanticscholar.org/CorpusID:258157664>, accessed on 16.04.2024.

²⁶⁸ Cf. Tamsah, Mohamad-Hani et al. "ChatGPT-4 and the Global Burden of Disease Study: Advancing Personalized Healthcare Through Artificial Intelligence in Clinical and Translational Medicine." *Cureus* 15 (2023): n. pag., <https://api.semanticscholar.org/CorpusID:258863204>, accessed on 16.04.2024.

²⁶⁹ Ebd.

may greatly increase drug adherence rates and promote overall patient outcomes by providing ongoing support and resolving issues.²⁷⁰

Furthermore, Chat GPT may provide valuable support to healthcare personnel in efficiently categorizing patients, particularly in high-traffic emergency rooms.²⁷¹

Chat GPT may efficiently select cases based on severity by gathering initial information from patients, allowing healthcare practitioners can deploy their limited resources more effectively. This application has the potential to reduce waiting times, speed up essential treatments, and improve overall patient satisfaction in the healthcare system.²⁷²

To summarize, while this bachelor thesis emphasizes the function of Chat GPT in chronic illness management, it is important to acknowledge the wide potential of this technology in several other healthcare domains. Chat GPT has the potential to transform healthcare services by offering assistance in mental health support, preventative care, education adherence, and triaging.²⁷³ Through the use of AI and NLP healthcare practitioners may provide tailored easily accessible, and efficient treatment to a wider audience, leading to enhanced patient outcomes and a complete transformation of the healthcare industry.²⁷⁴

9. Conclusion

The potential of Chat GPT technology to transform chronic illness management is becoming more clear in constantly changing field of healthcare. This subchapter seeks to provide a thorough overview of the main discoveries and understandings presented in this bachelor thesis “Unleashing the Potential of Chat GPT: Enhancing Chronic Disease Management in Healthcare Sector” with a specific focus on healthcare industry and the specialized field of chronic illness management.

One of the main conclusions is that Chat GPT has the capacity to assist persons with chronic diseases by providing tailored guidance. Using sophisticated NLP skills, Chat GPT is able to comprehend the distinct requirements and worries of

²⁷⁰ Cf. Kirsch, Verena and Jan Matthes. “Aspects of Medication and Patient participation—an Easy guideLine (AMPEL). A conversation guide increases patients’ and physicians’ satisfaction with prescription talks.” *Naunyn-Schmiedeberg's Archives of Pharmacology* 394 (2021): 1757 – 1767, <https://api.semanticscholar.org/CorpusID:235380233>, accessed on 16.04.2024.

²⁷¹ Cf. Jacob, J. Abel Joe et al. “Leveraging chatGPT for improved patient outcomes in a busy emergency department.” *International journal of health sciences* (2023): n. pag., <https://doi.org/10.53730/ijhs.v7nS1.14399>, accessed on 16.04.2024.

²⁷² Cf. Zhu, Zhoule et al. “ChatGPT’s potential role in non-English-speaking outpatient clinic settings.” *Digital Health* 9 (2023): n. pag., <https://api.semanticscholar.org/CorpusID:259655797>, accessed on 16.04.2024.

²⁷³ Cf. Islam, Md. Rabiul et al. “Role of ChatGPT in health science and research: A correspondence addressing potential application.” *Health Science Reports* 6 (2023): n. pag., <https://doi.org/10.1002/hsr2.1625>, accessed on 16.04.2024.

²⁷⁴ Ebd.

patients offering them personalized suggestions for effectively treating medical problems. This individualized approach enables patients to actively participate in their own healthcare journey, resulting in enhanced results and an improved quality of life.

Another significant revelation is that Chat GPT might have a pivotal function in monitoring symptoms. Chat GPT utilizes real-time data input from patients, including self-reported symptoms and vital signs, to analyze and identify trends. This enables it to promptly notify healthcare practitioners of impending exacerbations or problems. This proactive surveillance system not only improves patient safety but also facilitates early intervention, therefore averting acute episodes and decreasing healthcare expenses.

Moreover, Chat GPT has the capacity to provide lifestyle suggestions to persons suffering from chronic diseases. Chat GPT may provide tailored recommendations for food, exercise, stress management, and other lifestyle adjustments by taking into account a patient's medical history, present condition, and lifestyle characteristics. By providing these ideas, along with continuous support and encouragement, patients may be empowered to make constructive changes and enhance their overall state of well-being.

To summarize, this bachelor thesis emphasizes the significant capacity of Chat GPT to improve chronic illness management in the healthcare industry.

Chat GPT may enhance the well-being of persons with chronic diseases by offering tailored guidance, tracking symptoms, and suggesting lifestyle modifications, resulting in improved results and enhanced quality of life. With the ongoing advancement and integration of technology into healthcare systems, it's imperative for healthcare professionals and stakeholders to actively adopt and investigate the potential of Chat GPT in order to fully use its capabilities in managing chronic diseases.

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Affidavit

I declare that I have written the above thesis independently and have not made use of any outside help. I have identified as such all passages that are taken verbatim or in spirit from published or unpublished sources. The thesis has not been submitted in the same or similar form to any other examination authority.

Jilich 28.08.2024
place, date

H. Pelychraieka
signature

